U21BM603- Biomechanics

By

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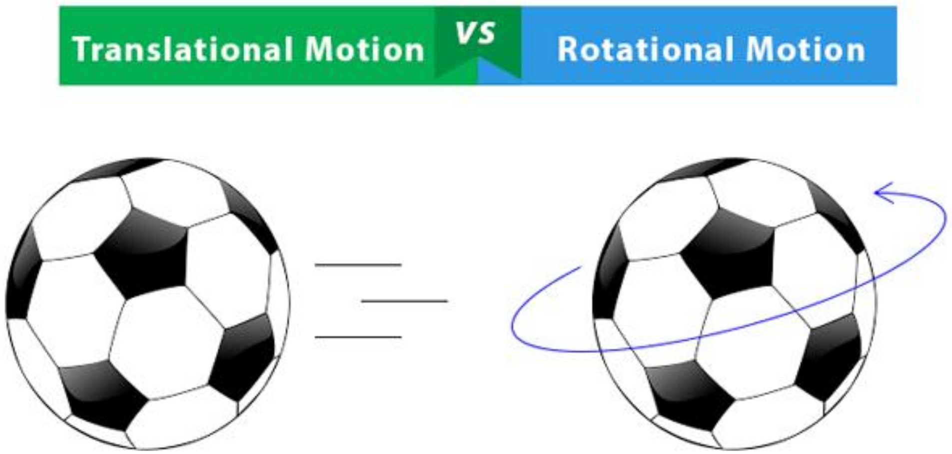
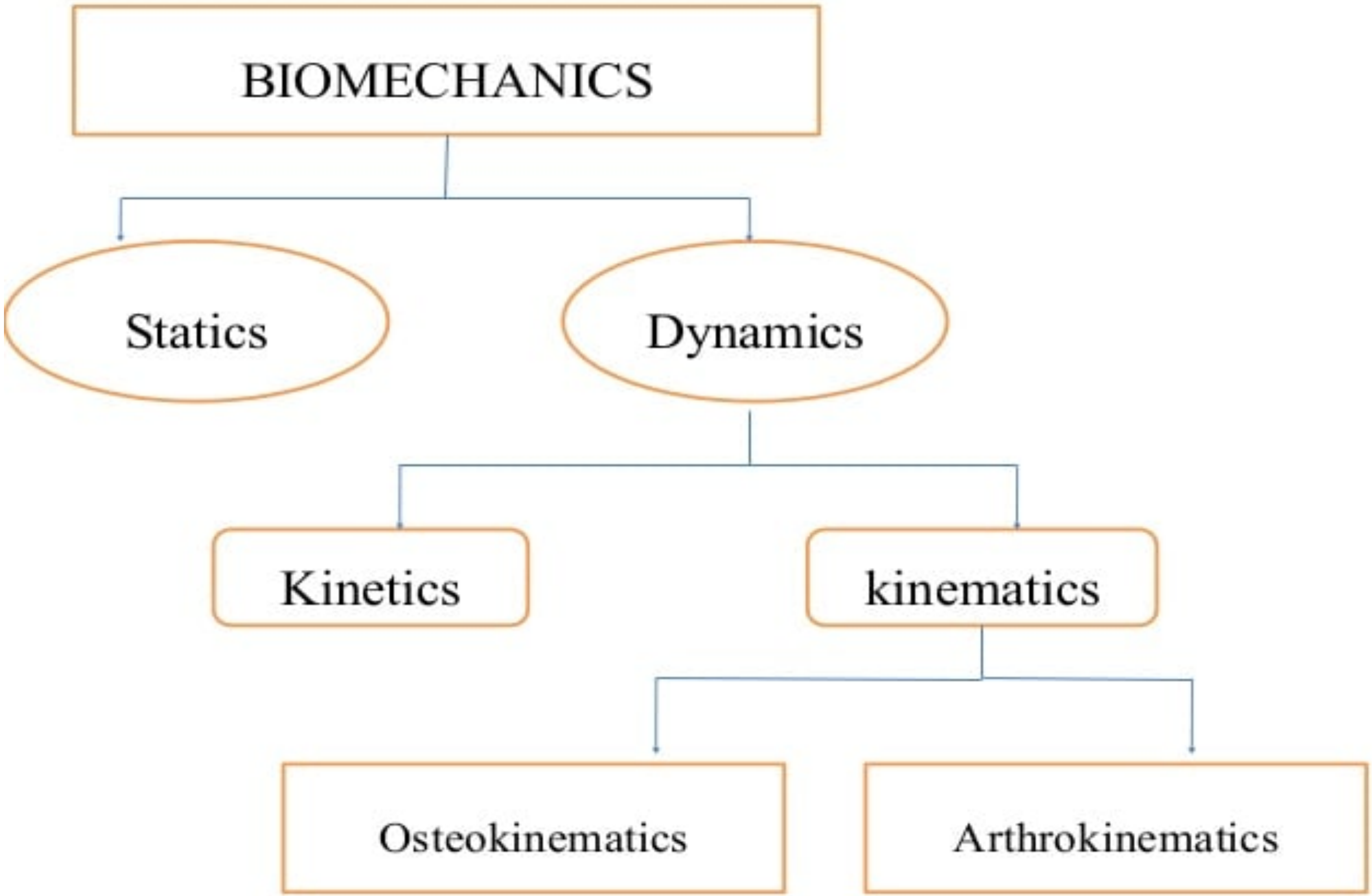
OBJECTIVES

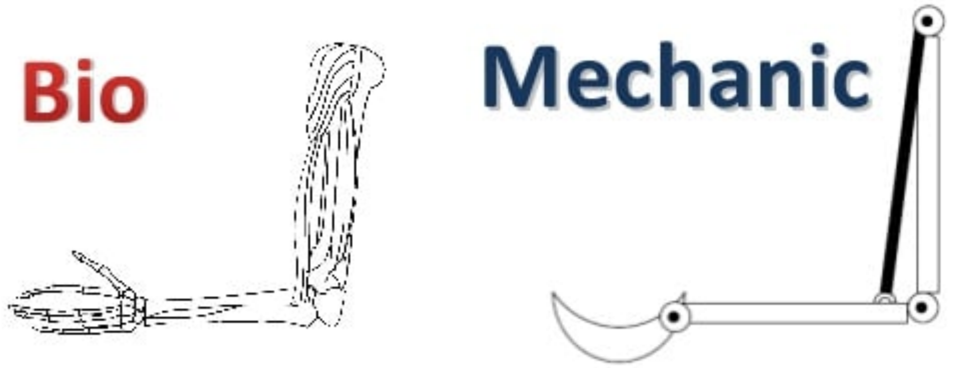
- Introduction to biomechanics
- Kinetics
- Types of forces
  - Inertial forces
  - Work, energy and power
  - Friction
- Kinematics
  - Rotational and translational motion
  - Osteokinemetics and Arthrokinematics




BIOMECHANICS
Statics
Dynamics
Kinetics
kinematics
Osteokinematics
Arthrokinematics

What is biomechanics?

- The term biomechanics combines the prefix bio, meaning “life,” with the field of mechanics, which is the study of the actions of forces.
- In biomechanics we analyze the mechanical aspects of living organisms.



- It is the study of the mechanics as it relates to the functional and anatomical analysis of the human body.

- Statics: involves all forces acting on the body being in balance, resulting in the body being in equilibrium.

- Dynamics: involves the study of systems in motion while unbalance due to unequal forces acting on the body.

Mechanics

Dynamics-moving systems

1. Kinetics

Examines the forces acting on the body during movement and the motion with respect to time and forces

2. Kinematics

A branch of biomechanics that describes the motion of a body without regard to the forces that produce the motion

What is kinetics?

- Kinetics is the study of motion under the action of forces.
- Forces that cause, arrest, or modify moti on in a system
  - Gravity
  - Muscles
  - Friction
  - External resistance

Effect of forces on the body

- The Force -

Any action or influence that moves a body or influences the movement of a body

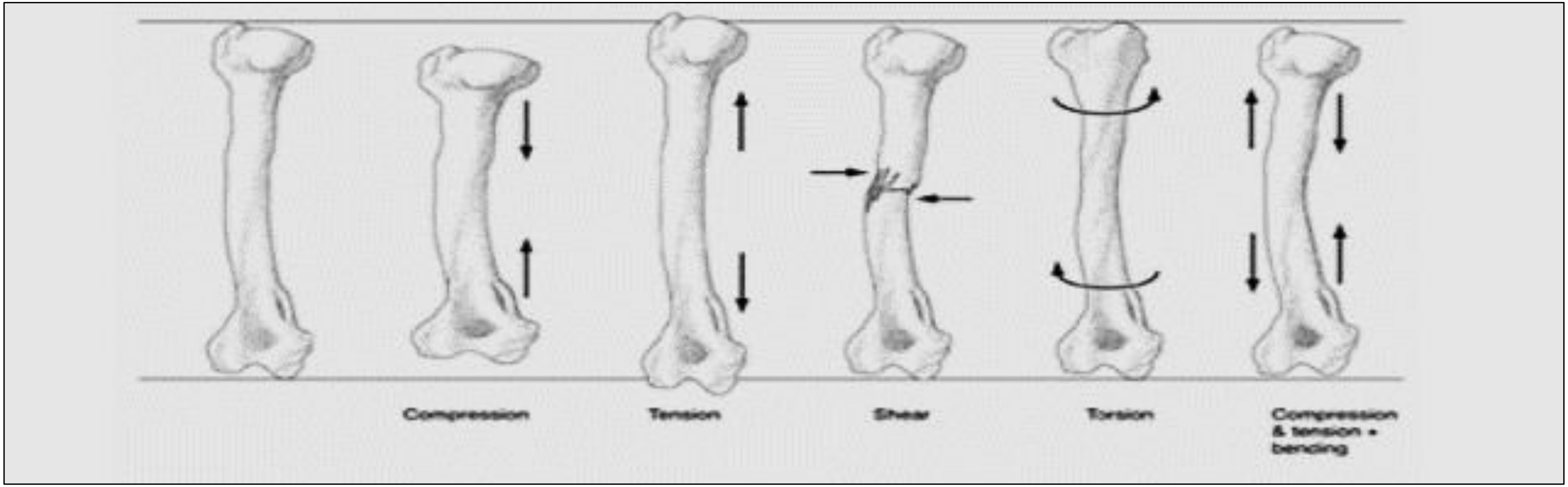
- Forces "control" the movement of the body

INTERNAL	EXTERNAL
Muscle contraction	Gravity
Tension from ligaments	An external load
Muscle lengthening	A therapist applying resistance or a free-weight for resistance training

Force

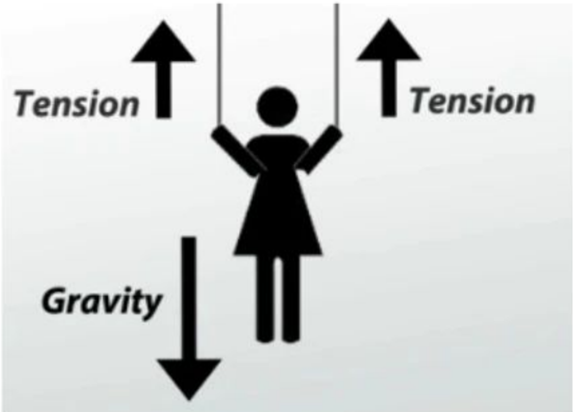
- A force is describe as a push or pull exerted by one object or substance on another.
- Whether a body or body segments is in motion or at rest depends on the forces exerted on that body.
- Any two objects make contact, they will either push on each other or pull on each other with some magnitude of forces.
- Force = (mass)(acceleration)
- The unit of force in the SI unit is the newton (N)

- Excessive tissue deformation due to mechanical loading may result from
  - Tension (stretching or strain)
  - Compression and distraction
  - Shear
  - Bending
  - Torsion (twisting)



Tension

- Tension stress (or tensile stress) occurs when two forces pull an object in opposite directions so as to stretch it and make it longer and thinner.
- The primary load a muscle experiences is a tension load.
- When the muscle contracts it pulls on the tendons at both ends, which stretch a little. So, the tendons are under tensile stress.



Compression force

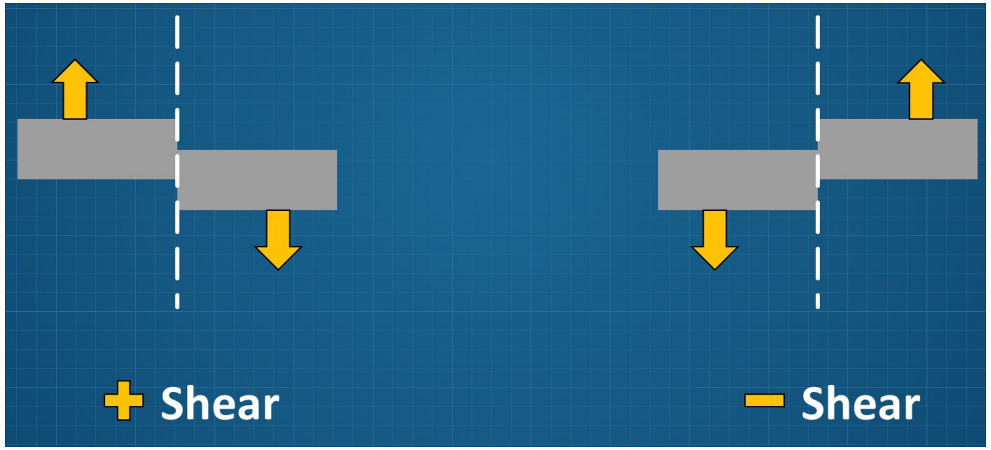
- Compression pushes or presses an object so as to make it shorter and thicker.
- Tension and compression stress are both sometimes referred to as axial stress because the forces act along a structure's longitudinal axis.

Distraction force

- A net force that moves a bony segment away form its adjacent bony segment is known as distraction force.
- A distraction force tends to cause a separation between the bones that make up a joint .

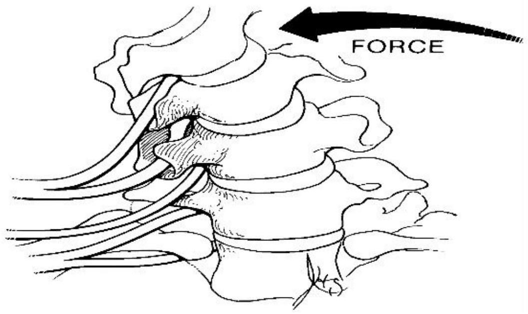
Shear force

- Shear force is two forces acting parallel to each other but in opposite directions so that one part of the object is moved or displaced relative to another part.
- Shear causes two objects to slide over one another this results, in friction.
- If one vertebra is being caused to slide relative to another then there is a shear between them.



Bending and Torsion

- Bending is a loading mode results in the generation of maximum tensile forces on the convex surface of the bent member and maximum compressive forces on the concave side.
- Torsion is when forces acting on a structure cause a twist about its longitudinal axis.



Torque

- The strength of rotation produced by a force couple is known as torque.
- The internal and external forces acting at a joint
- The rotational equivalent of force
- Torque(τ) = (D) x (F)

- Mechanics :
- Mass
  - Amount of matter that a body contains.
- Inertia
  - Property of matter that causes it to resist any change of its motion in either speed or direction.

Inertial Forces

- Kinematics and kinetics are bound by Newton’s second law, which states that the external force (f) on an object is proportional to the product of that object’s mass (m) and linear acceleration (a): $f = ma$
- Mass is a measure of inertia.
- Resistance to a change in motion.

Work, Energy, and Power

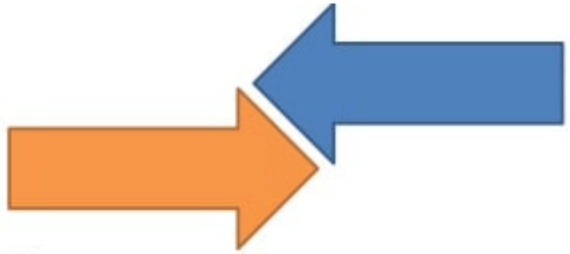
- Work is defined as the force required to move an object a certain distance

(work = force x distance)

- The standard unit of work in the metric system is a joule(J; newton x meter).
- Power is defined as the rate of work is being done (power = work/time).
- The standard unit of power is a watt (W; watt=newton meter/second).
- The energy of a system refers to its capacity to perform work.
- Energy has the same unit as work (J) and can be divided into potential and kinetic energy.
- While potential energy refers to stored energy, kinetic energy is the energy of motion.

Friction

- A force that is developed by two surfaces.
- Frictional forces can prevent the motion of an object when it is at rest and resist the movement of an object when it is in motion.
- The values for the coefficient of friction depend on several parameters, such as the composition and roughness of the two surfaces in contact.



What is kinematics

- kinematics is defined as the study of motion without regard to the forces that cause that motion.
- Involves the time, space, and mass aspects of a moving system.
- The description of motion :
  - Osteokinematics: the manner in which bones move
  - Arthrokinematics: movements occurring between joint surfaces.

Kinematics - Types of Motion

- Translatory - Movement of a body in which all of its parts move in the same direction and distance and at the same speed

    1. Rectilinear motion = straight line motions (sliding surfaces)

    2. Curvilinear motion = curved line of motion (the motion of a ball when tossed)

Rotatory motion

- The arc of motion around a fixed axis of rotation or a "pivot point" .
- Joints have "pivot points" which are used as reference points from which to measure the range of motion (ROM) of that joint.

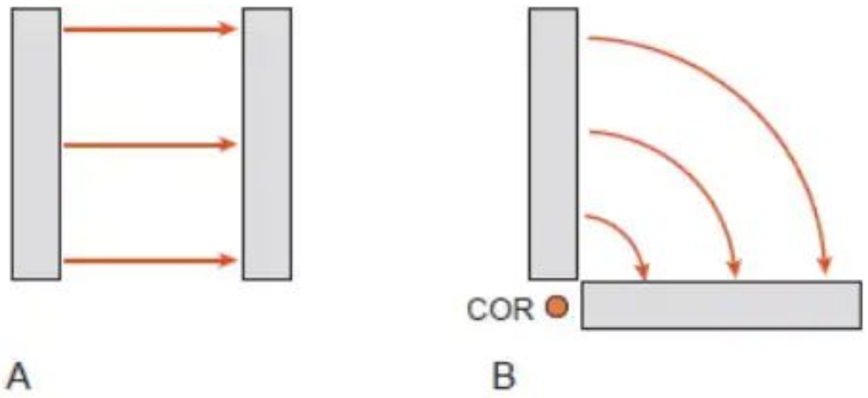


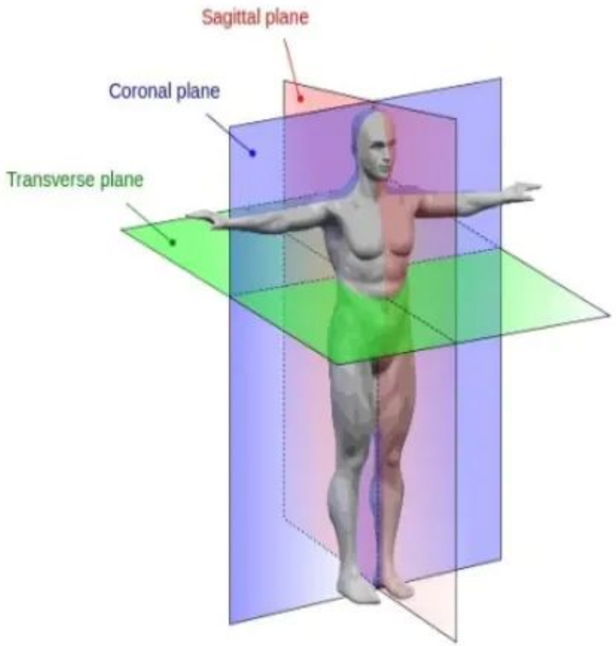
Figure 1.11: Translations and rotations. In biomechanics, motion is typically described in terms of translations and rotations. **A.** In translatory motion, all points on the object move the same distance. **B.** In rotational motion, all points on the object revolve around the center of rotation (*COR*), which is fixed in space.

Osteokinematics

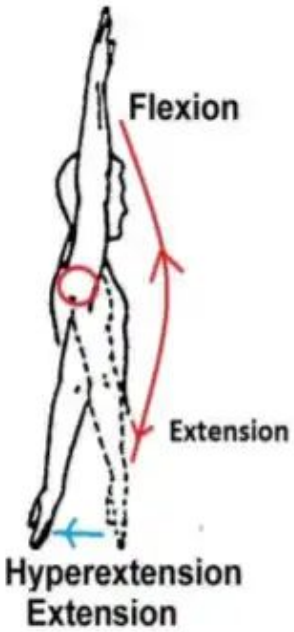
- To define joint and segment motions and to record the location in space of specific points on the body, a reference point is required.
- The three dimensional rectangular coordinate system is used to describe the relationship of the body. The origin of x-axis, y-axis and z-axis of the coordinate system is traditionally located at the center of mass (CoM) of the body.

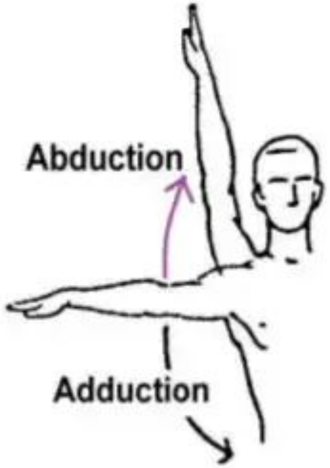
- The option for movements of a segment are also referred to as degrees of freedom.
- Three imaginary planes are arranged perpendicular to each other through the body, with their axes intersecting at the center of gravity of the body.
- These planes are called the cardinal planes of the body.

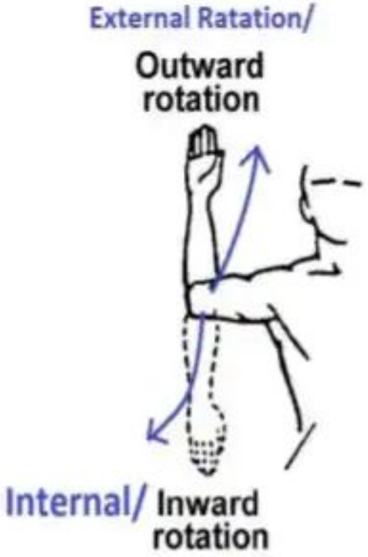
- Motion of bones through a range of motion relative to the 3 cardinal planesof the body and around the axis in that joint
- Planes:
  - Frontal or Coronal
  - Sagittal or Median
  - Horizontal or Transverse

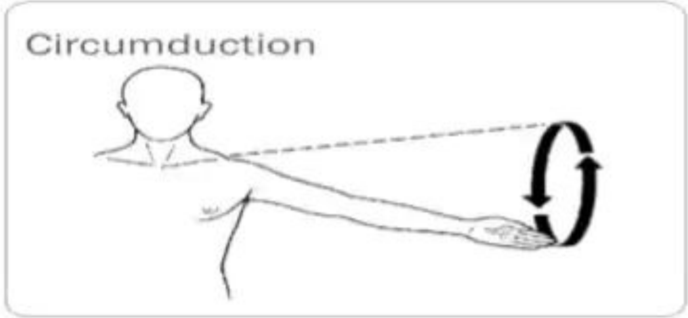


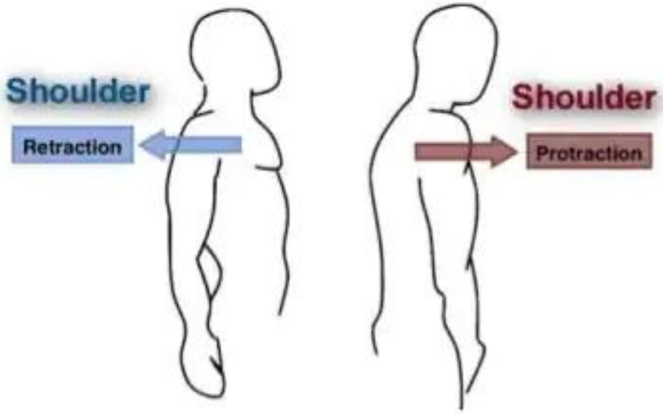
Osteokinematics: Fundamental Motions

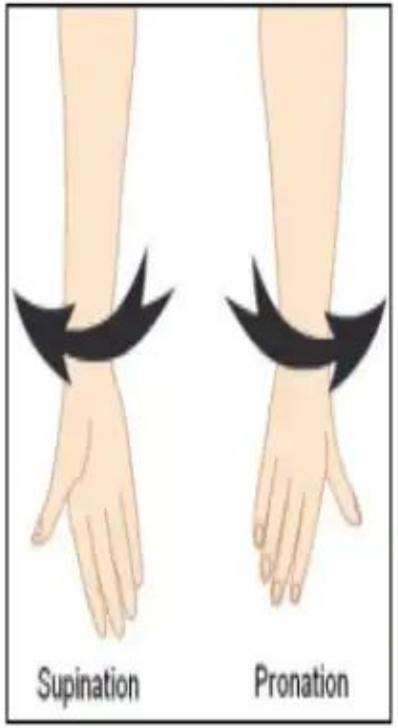


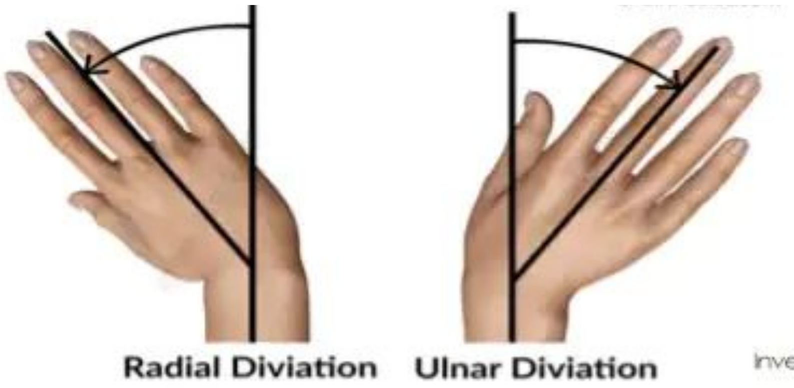




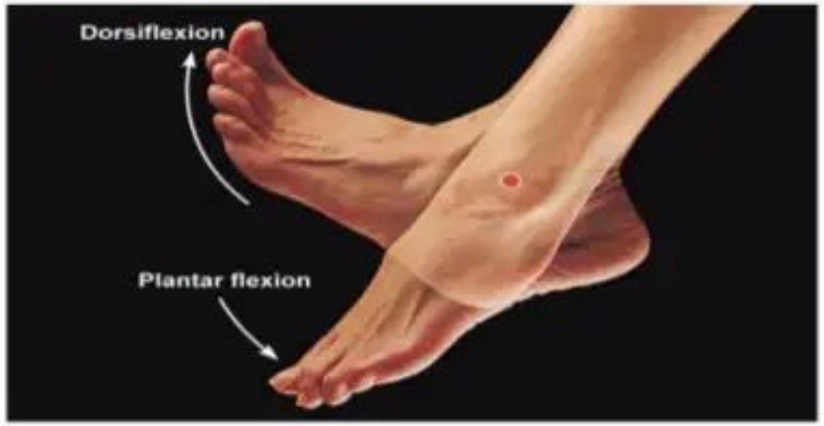







Radial Diviation
Ulnar Diviation


Inversion
Eversion

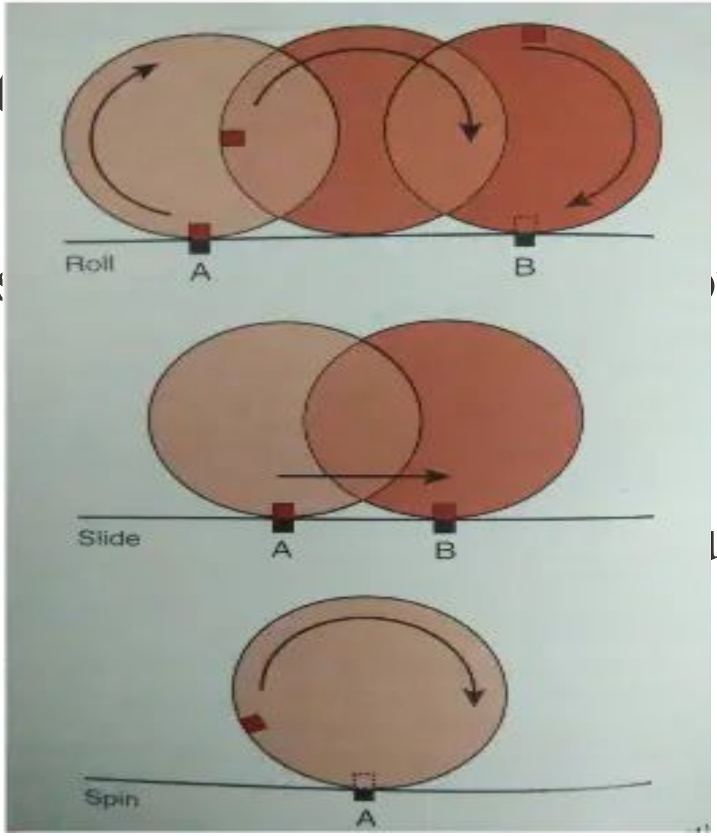

Dorsiflexion
Plantar flexion

Arthrokinematics

- Manner in which adjoining joint surfaces move in relation to each other or how they fit together
- helps to improve the movement of the joint
- Parts may move in:
  - The same direction
  - The opposite direction

Fundamental Movements: Joint Surfaces

- Roll
  - Multiple points maintain contact throughout
- Slide
  - A single point on one surface contacts ut the motion
- Spin
  - A single point on one surface rotates on a urface

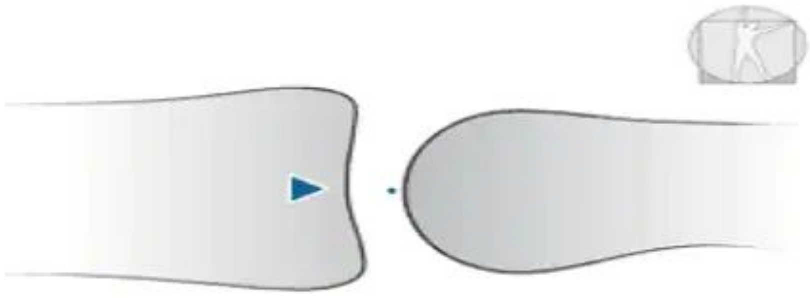


Roll & Slide Mechanics

• Convex on Concave

 When a convex joint surface moves on a concave joint surface

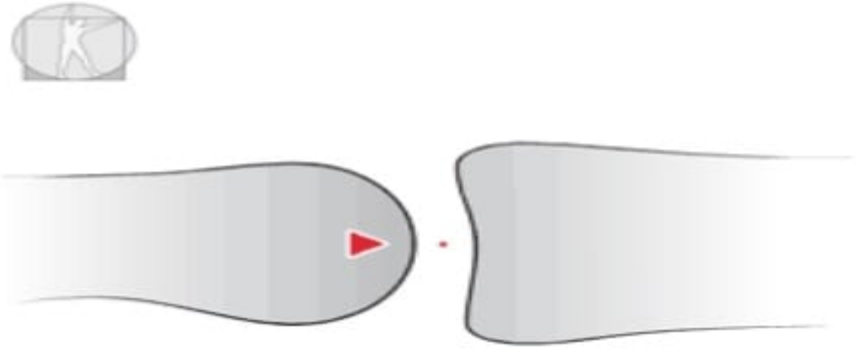
 The roll and slide occur in opposite directions



- Concave on Convex

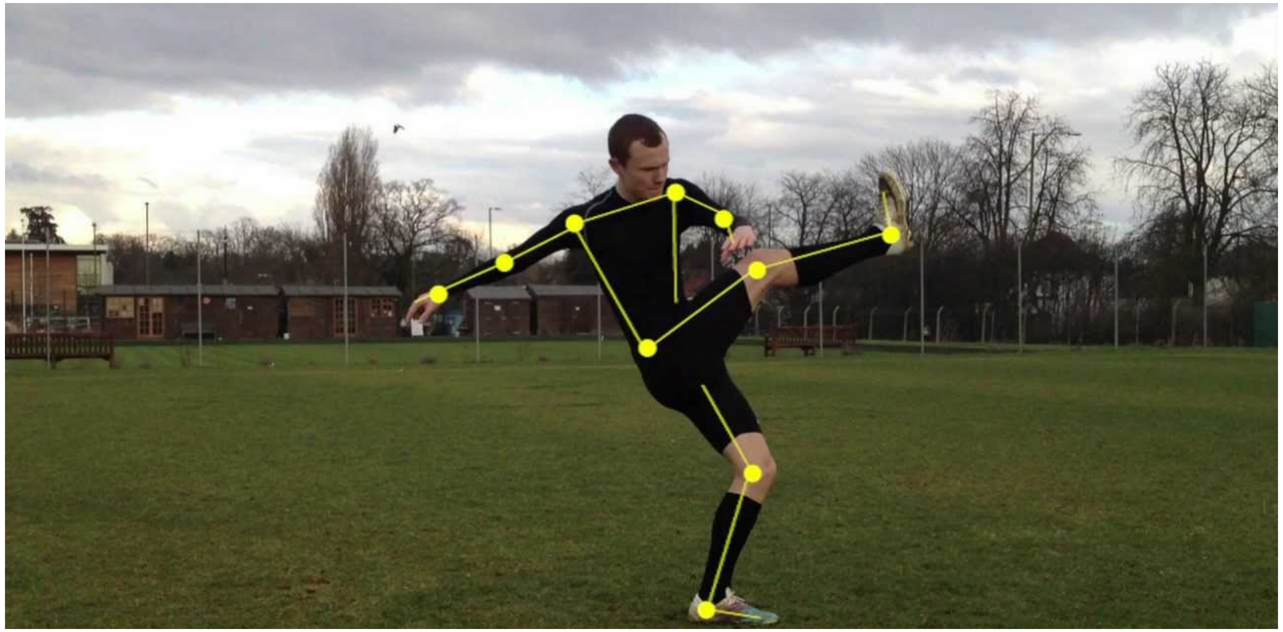
 When a concave joint surface moves about a stationary convex joint surface

  the roll and slide occur in the same direction



Case study

A professional football team sought to enhance its players' performance and reduce the risk of injuries by incorporating kinematic analysis into their training regimen. The goal was to use biomechanical insights to refine players' movements, improve overall agility, and optimize on-field performance.



PROBLEMS

1. A 58-kg gymnast is performing a giant swing. The velocity of her center of mass is 1.3 m/s. Her height is 3.7 m. Her body is stretched 11 cm with a stiffness of 5 kN/m. What is:

a. Her kinetic energy

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}(58)(1.3^2) = 49\ \text{J}$$

b. Her gravitational potential energy

$$GPE = -mgh = -58(-9.8)(3.7) = 2103\ \text{J}$$

c. Her strain potential energy

$$SPE = \frac{1}{2}k\Delta x^2 = \frac{1}{2}(5000)(0.11^2) = 30\ \text{J}$$

d. Her total mechanical energy

$$E = KE + GPE + SPE = 49 + 2103 + 30 = 2182\ \text{J}$$

2. A 0.15-kg baseball strikes the catcher's glove with a horizontal velocity of 40 m/s. The displacement of the baseball due to the deformation of the catcher's glove and the movement of the catcher's hand is 8 cm in a horizontal direction from the instant it first makes contact with the glove until it stops.

a. How much kinetic energy does the baseball possess just before it strikes the glove?

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.15)(40^2) = 120\ \text{J}$$

b. How much work does the catcher do on the baseball during the catch?

Since work is equal to the change in energy and there is not change in height or strain energy, then $U = \Delta E = \Delta KE = -120\ \text{J}$

c. What is the average impact force exerted by the glove on the baseball?

$$U = Fd$$

$$F = \frac{U}{d}$$

$$F = \frac{-120}{-0.08} = \boxed{1500\text{ N}}$$

d. Is the work done positive or negative?

The work is negative since energy is decreasing.

3. Nick generates a horizontal release velocity of 18 m/s on a 2.0 kg discus. If he applies force to the disc for 0.58 s, what is the average horizontal power the thrower exerted on the disc?

Step 1: $KE = \frac{1}{2}mv^2 = \frac{1}{2}(2)(18^2) = 324\ \text{J}$

Step 2: Since we are only considering the horizontal direction:

$$U = \Delta E = \Delta KE$$

$$P = \frac{U}{t} = \frac{324J}{0.58} = 559\ \text{W}$$

4. The coefficient of static friction between a tennis player's hand and her racket is 0.45. How hard must she squeeze the racket if she wants to exert a force of 200 N along its longitudinal axis?

$F = \mu N$

$F = 200N$

$\mu = 0.45$

$200N = 0.45(\text{Normal Force})$

$\text{Normal force} = 200N/0.45 =$ **444N**

5. The coefficient of static friction between the sole of an athletic shoe and the basketball court floor is 0.67. Tyler wears these shoes when he plays basketball. If Tyler exerts a normal contact force of 1400 N when he pushes off the floor to run down the court, how large is the friction force exerted by Tyler's shoes on the floor?

Frictional force = F
Coefficient of static friction = 0.67
Normal force = 1400N

$F = \mu N$
$F = 0.67\ (1400N) = 938N$

Joints

- Articulations of bones
- Functions of joints
    - Hold bones together securely
    - Gives the rigid skeleton mobility
- Ways joints are classified
    - Functionally
    - Structurally

Functional Classification of Joints

- Focuses on the amount of movement allowed
- Joint types restricted mainly to the axial skeleton where firm attachments and protection of internal organs are priorities include:
  - Synarthroses – immovable joints
  - Amphiarthroses – slightly moveable joints
- Joints predominate in the limbs where mobility is important:
  - Diarthroses – freely moveable joints

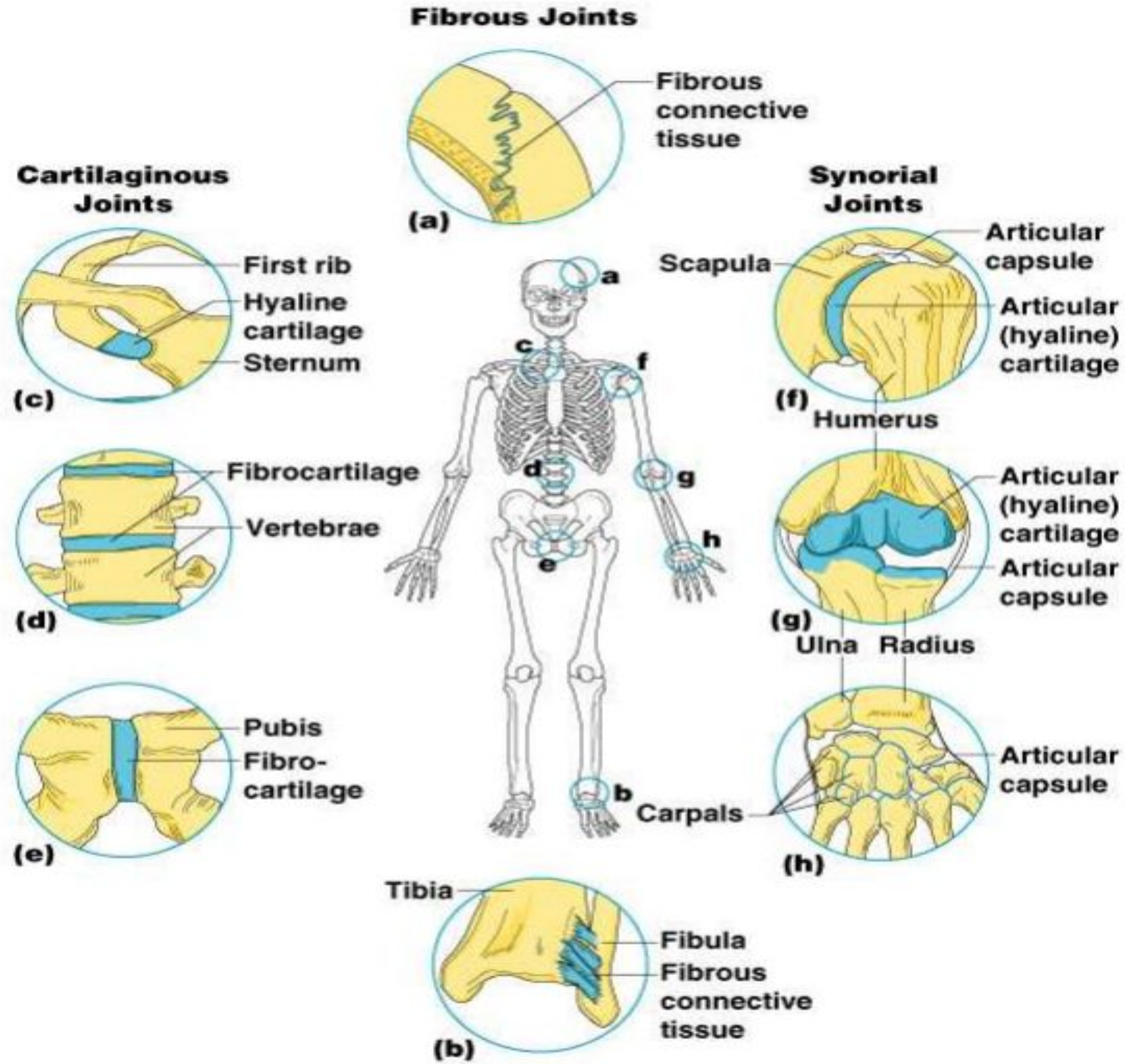
Structural Classification of Joints

- Based on whether fibrous tissue, cartilage or a joint cavity separates the bony regions.

  General rule:
- Fibrous joints
  - Generally immovable
- Cartilaginous joints
  - Some are immovable (synarthrotic), while most are slightly moveable (amphiarthrotic)
- Synovial joints
  - Freely moveable (diarthrotic)

Fibrous Joints

- Bones united by fibrous tissue
- Sutures – irregular edges of bone interlock, bound tightly by connective tissue fibers = essentially no movement
- Syndesmoses – connecting fibers are longer and have more give
- Example: joint connecting distal end of tibia and fibula


Fibrous Joints
Fibrous
connective
tissue
(a)
Cartilaginous
Joints
First rib
Hyaline
cartilage
Sternum
(c)
Fibrocartilage
Vertebrae
(d)
Pubis
Fibro-
cartilage
(e)
Synorial
Joints
Scapula
Articular
capsule
Articular
(hyaline)
cartilage
(f)
Humerus
Articular
(hyaline)
cartilage
Articular
capsule
(g)
Ulna
Radius
Articular
capsule
Carpals
(h)
a
c
f
d
g
h
e
b
Tibia
Fibula
Fibrous
connective
tissue
(b)

Cartilaginous Joints

- Bones ends connected by cartilage
- Slightly moveable (amphiarthrotic)
  - o Pubic symphysis
  - o Intervertebral joints
- Immovable (synarthrotic)
  - o Hyaline-cartilage epiphyseal plates of growing long bones
  - o Cartilaginous joints between the 1st ribs and the sternum

Synovial Joints

- Articulating bones are separated by a joint cavity
- Synovial fluid is found in the joint cavity
- All joints of the limbs

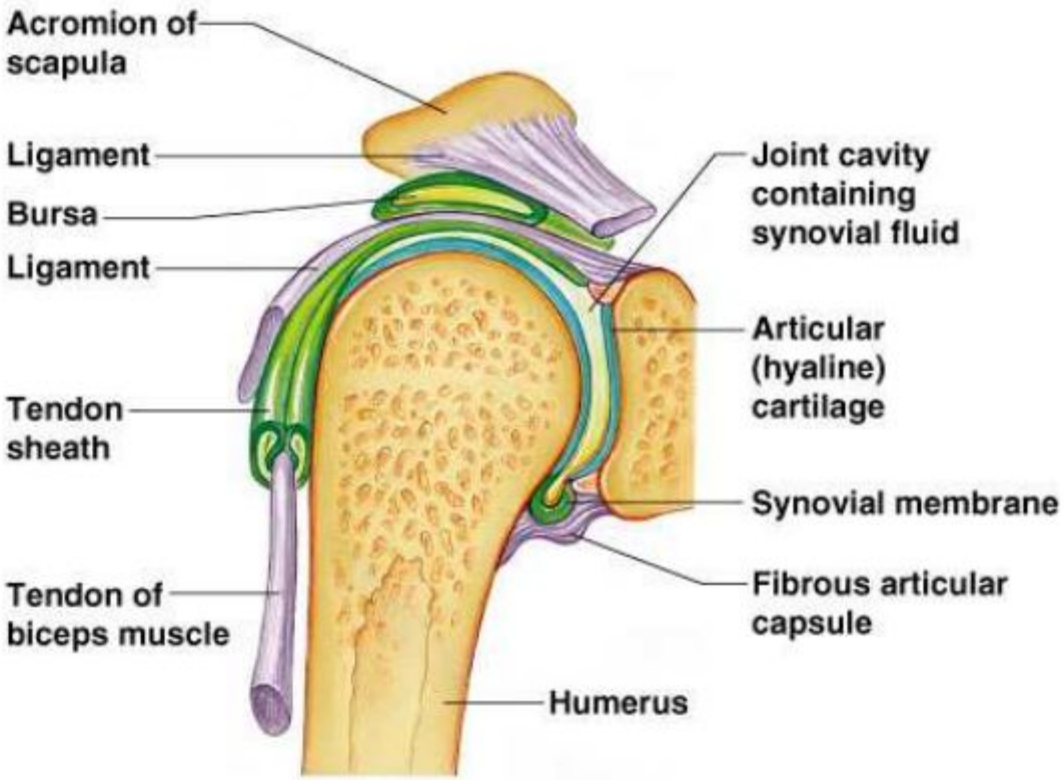
Features of Synovial Joints

- Articular cartilage (hyaline cartilage) covers the ends of bones forming the joint
- A fibrous articular capsule lined with a smooth synovial membrane encloses joint surfaces
- Have a joint cavity filled with lubricating synovial fluid
- Ligaments surround the capsule and reinforce the joint

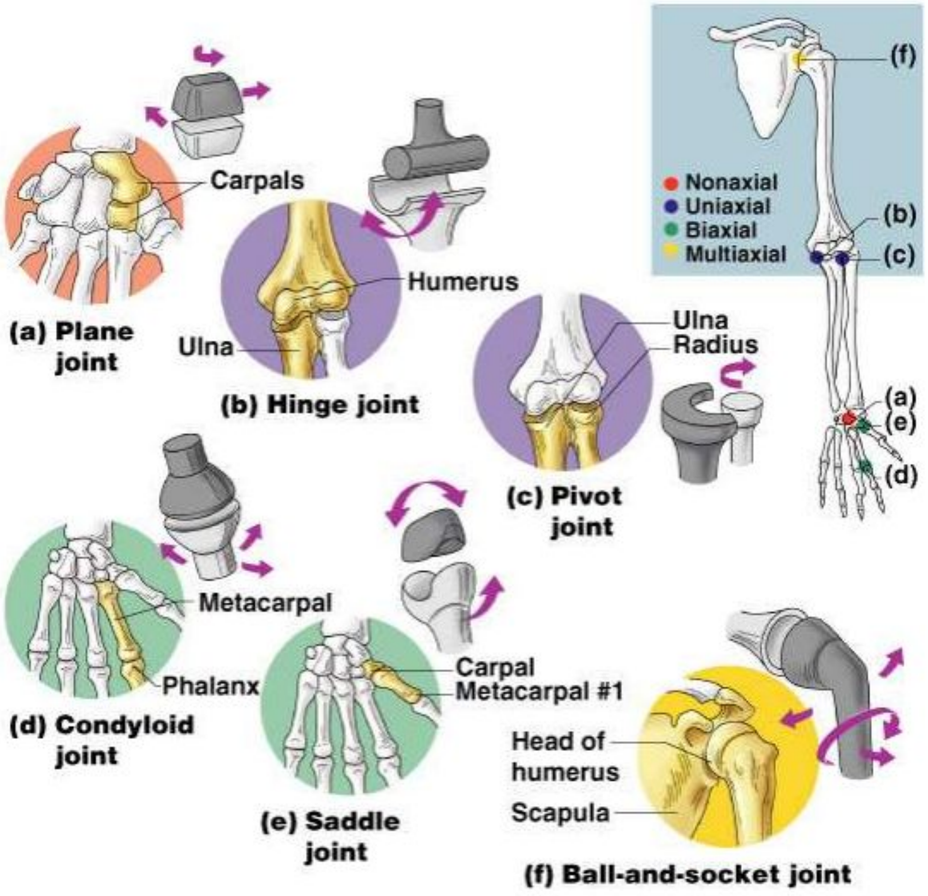
Structures Associated with the Synovial Joint

- Bursae – flattened fibrous sacs
- Lined with synovial membranes
- Filled with synovial fluid
- Not actually part of the joint – act like ball bearings to reduce friction
- Common where ligaments, muscles, skin, tendons or bones rub together

The Synovial Joint



Types of Synovial Joints based on shape



- The shapes of the articulating bone surfaces determine the movement of a joint.
- Plane joint
  - Articular surfaces are essentially flat & only short, gliding movements are
  - allowed.
  - Nonaxial = no rotation.
  - Ex: intercarpal joints of wrist.
- Hinge joint
  - Cylindrical end of 1 bone fits into a trough-shaped surface of another
  - bone.
  - Angular movement in one plane.
  - Uniaxial = allow movement around one axis only.
  - Ex: elbow, ankle, & joints between phalanges.

- Pivot joint
  - Rounded end of 1 bone fits into a sleeve or ring of bone (& possibly ligaments).
  - Uniaxial – can only turn around its long axis.
  - Ex: proximal radioulnar joint & joint between the atlas & the dens of the axis.
- Condyloid joint/ A.K.A Ellipsoid joint (knuckle-like)
  - Egg-shaped articular surface of 1 bone fits into the oval concavity of another.
  - Both surfaces are oval.
  - Allows the moving bone to travel from side to side & back and forth, but can’t rotate around its long axis.
  - Biaxial (movement around 2 axes)
  - Ex: metacarpophalangeal joints (knuckles)

- Saddle joint
  - Each articular surface has both convex & concave areas, like a saddle Biaxial like the condyloid joints
  - Ex: carpometacarpal joints in the thumb (twiddling!)
- Ball & Socket joint
  - Spherical head of one bone fits into round socket in another Multiaxial joint (mov’t in all axes including rotation)
  - The most freely moving synovial joint
  - Ex: shoulder & hip

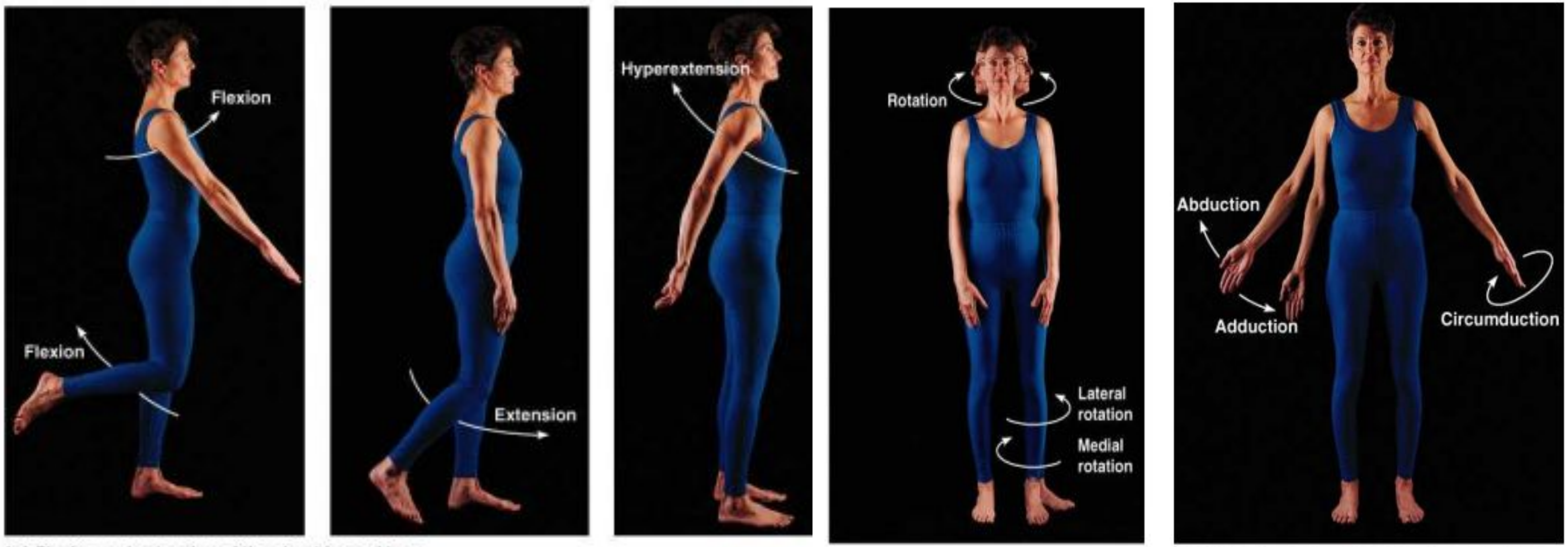
Inflammatory Conditions Associated with Joints

- Bursitis – inflammation of a bursa usually caused by a blow or friction
- Tendonitis – inflammation of tendon sheaths
- Arthritis – inflammatory or degenerative diseases of joints
  - Over 100 different types
  - The most widespread crippling disease in the United States

Body Movements

- Flexion
  - The process of bending or the state of being bent.
- Extension
  - The act of straightening or extending a flexed limb
- Hyperextension
  - Flexion of a limb or part beyond its normal range

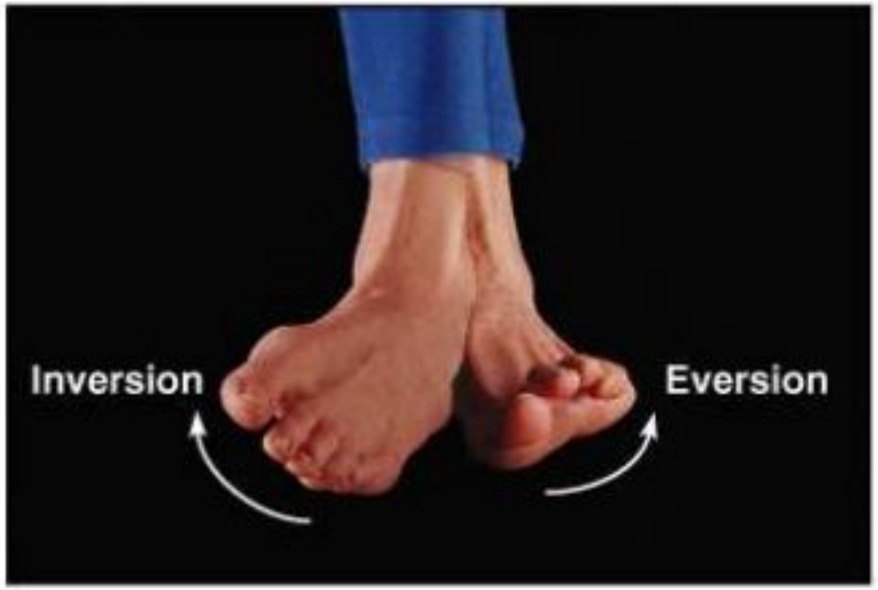
- Rotation
  - The process of turning around an axis.
- Abduction
  - To draw away from the median plane, or (the digits) from the axial line of a limb
- Adduction
  - To draw toward the median plane or (in the digits) toward the axial line of a limb
- Circumduction
  - Movement of a part in a circular direction



(a) Flexion and extension of the shoulder and knee
(c) Rotation
(d) Abduction, adduction, and circumduction

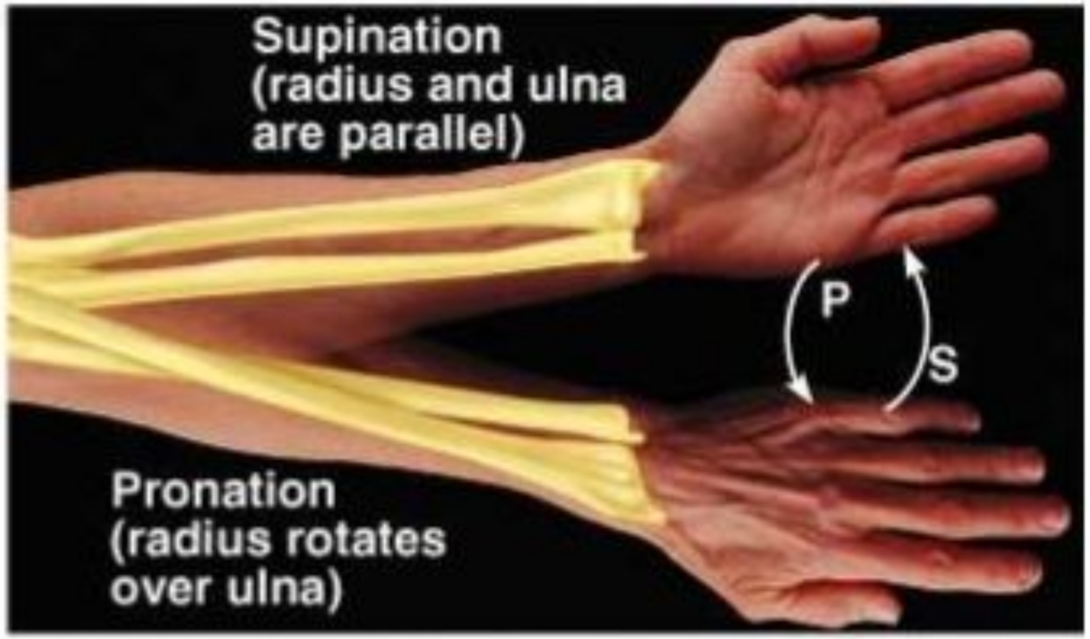
- Dorsiflexion
  - The turning of the foot or the toes upward
- Plantar flexion
  - Extension of the ankle resulting in the forefoot moving away from the body
- Inversion
  - Turning inward
- Eversion
  - Turning outward

(e) Dorsiflexion and plantar flexion



(f) Inversion and eversion

- Supination
  - Applied to the hand, the act of turning the palm forward (anteriorly) or upward, performed by lateral rotation of the forearm
  - Applied to the foot, it generally implies movements resulting in raising of the medial margin of the foot, hence of the longitudinal arch
- Pronation
  - Applied to the hand, the act of turning the palm backward (posteriorly) or downward, performed by medial rotation of the forearm.
  - Applied to the foot, a combination of eversion and abduction movements taking place in the tarsal and metatarsal joints and resulting in lowering of the medial margin of the foot, hence of the longitudinal arch



(g) Supination (S) and pronation (P)

- Opposition
  - The thumb, unlike other fingers, is opposable, in that it is the only digit on the human hand which is able to oppose or turn back against the other four fingers, and thus enables the hand to refine its grip to hold objects which it would be unable to do otherwise

(h) Opposition

Kinetic Concepts for Analyzing Human Motion

Basic Concepts Related to Kinetics

What is **mass**?

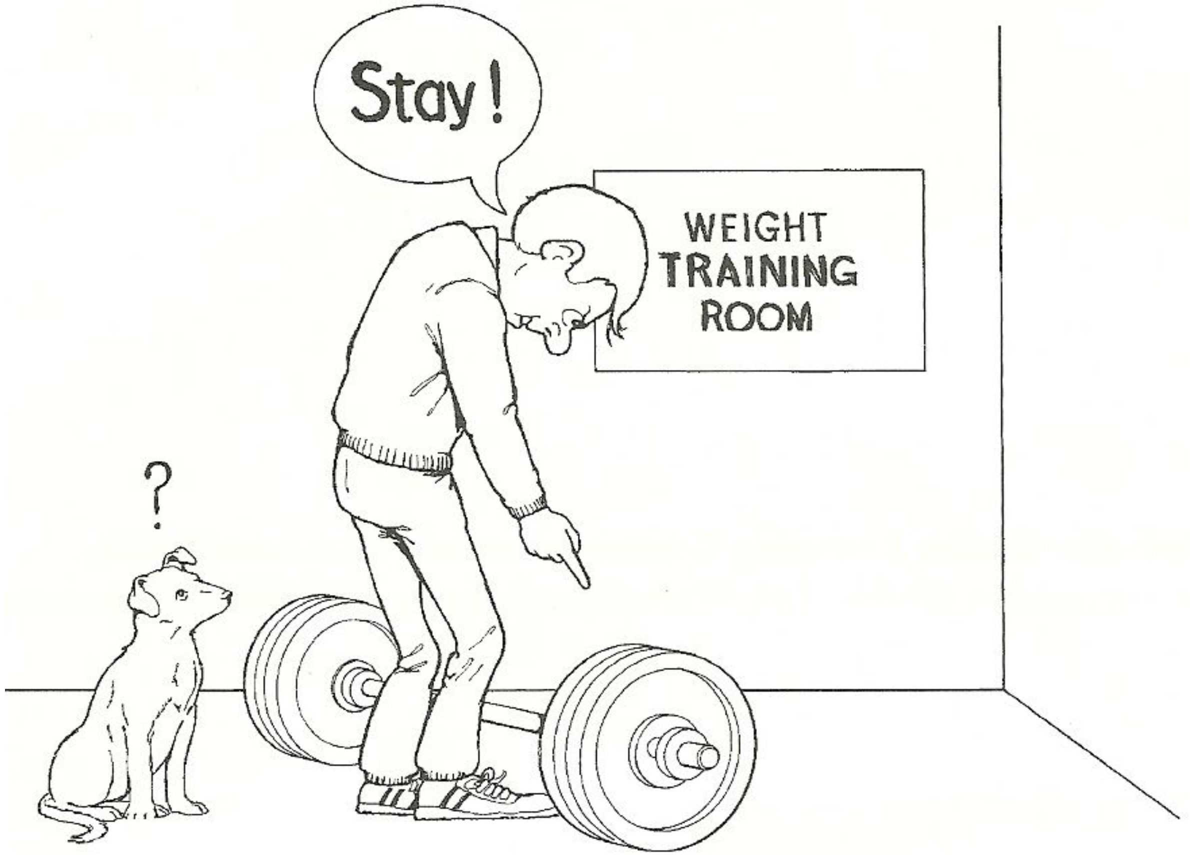
- Quantity of matter composing a body (dog, tree, desk, basketball, you)
- Represented by m
- Units are kg

Basic Concepts Related to Kinetics

What is **inertia**?

- Tendency to resist change in state of motion
- Proportional to mass
- Has no units!

Basic Concepts Related to Kinetics



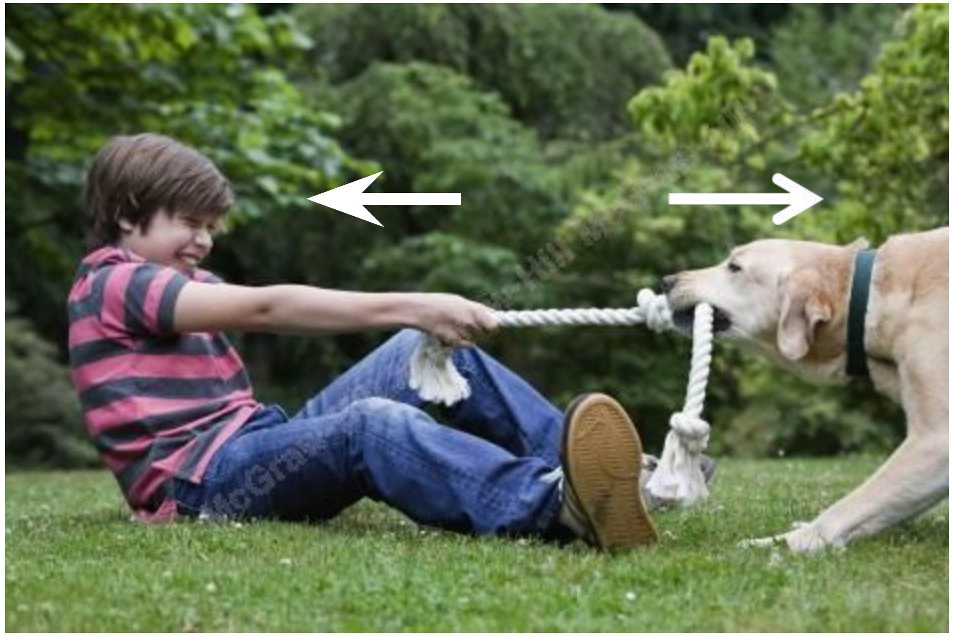
Clearly, the weight bar will stay in place in the absence of being lifted because of its inertia.

Basic Concepts Related to Kinetics

What is **force**?

- A push or a pull
  - Characterized by magnitude, direction, and point of application
- F = ma
- Unit is the Newton (N)

Basic Concepts Related to Kinetics

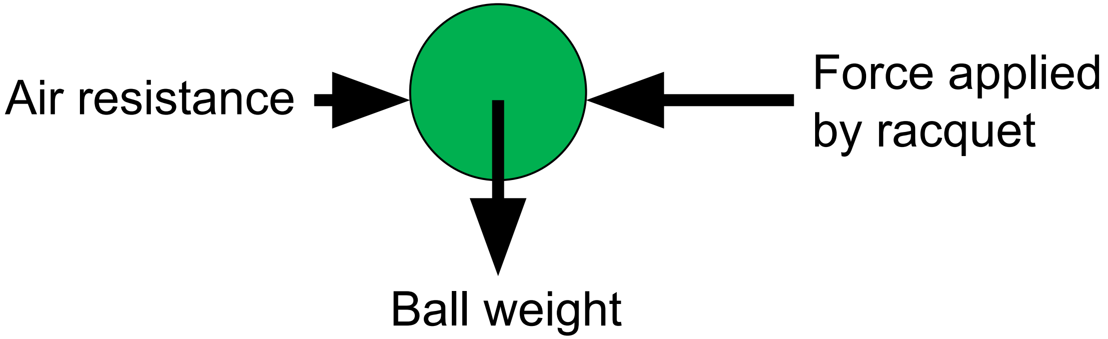


While there is no motion occurring, boy and dog are exerting equal and opposite forces on the rope.

Basic Concepts Related to Kinetics

What is a **free body diagram**?

Ball being struck by a racquet



(Diagram showing vector representations of all forces acting on a defined system)

Basic Concepts Related to Kinetics

What is a **net force**?

- The single resultant force derived from the vector composition of all the acting forces
- The force that determines the net effect of all acting forces on a body

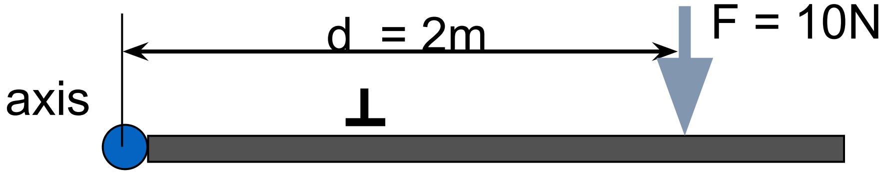
Basic Concepts Related to Kinetics

What is a **torque** (T)?

- The rotary effect of a force
- The angular equivalent of force
- Also known as moment of force

Basic Concepts Related to Kinetics

What is a **torque**?



$T = Fd_\perp$

$T = (10N)(2m)$

$T = 20\ Nm$

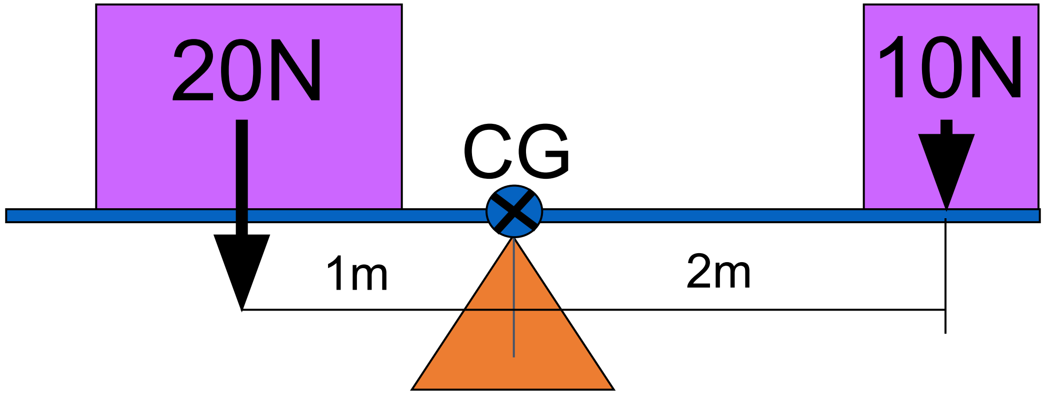
$T = Fd_\perp$ (the product of force and the perpendicular distance from the force's line of action to the axis of rotation)

Basic Concepts Related to Kinetics

What is the **center of gravity**?

- Point around which a body’s weight is equally balanced in all directions
- Point that serves as an index of total body motion
- Point at which the weight vector acts
- Same as the center of mass

Basic Concepts Related to Kinetics



The weights are balanced, creating equal torques on either side of the fulcrum.

Basic Concepts Related to Kinetics

If one child on a see-saw is heavier than the other, how can they balance?

Basic Concepts Related to Kinetics

What is **weight**?

- Attractive force that the earth exerts on a body
- wt. = ma_g (product of mass and the acceleration of gravity: $-9.81\ m/s^2$)

Basic Concepts Related to Kinetics

What is **weight**?

- The point of application of the weight force is a body's center of gravity
- Since weight is a force, units of weight are units of force: N

Basic Concepts Related to Kinetics

What is **pressure**?

- Force per unit of area over which the force acts
- Commonly used to describe force distribution within a fluid (e.g. blood pressure, water pressure)
- Units are N/m^2

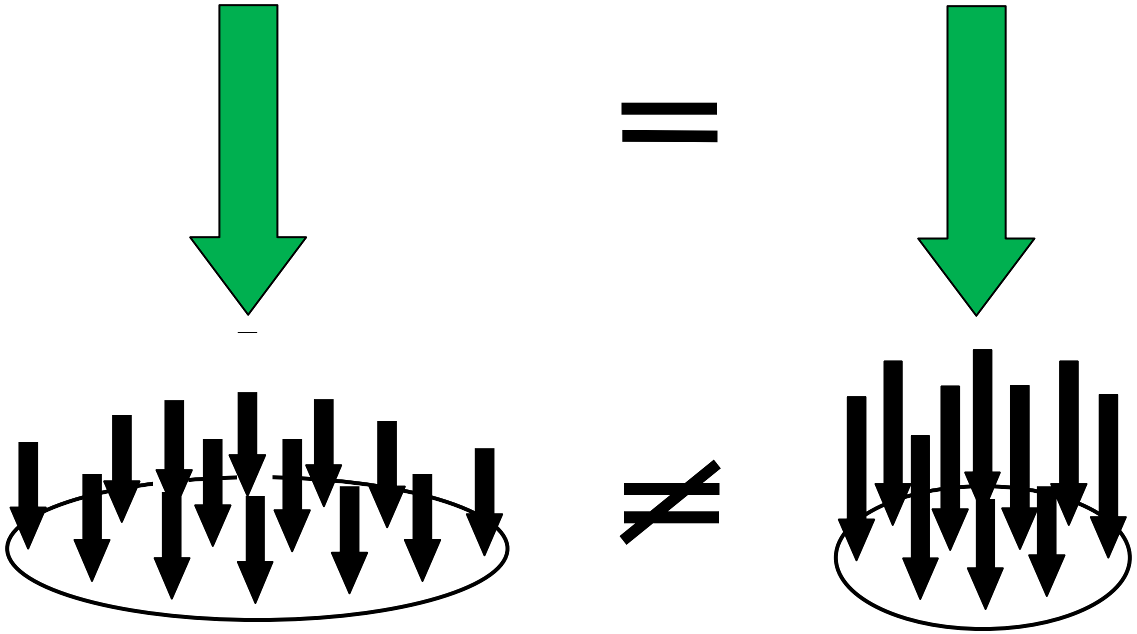
Basic Concepts Related to Kinetics

What is **stress**?

- Force per unit of area over which the force acts
- Commonly used to describe force distribution within a solid
- Units are N/m^2

Basic Concepts Related to Kinetics

What is **stress**?



Basic Concepts Related to Kinetics

Why would it hurt more to be stepped on by someone in high heels as compared to the same person wearing athletic shoes?

Basic Concepts Related to Kinetics

What is **volume**?

- Space occupied by a body
- Has three dimensions (width, height, and depth)
- Units are m^3 and cm^3

Basic Concepts Related to Kinetics

What is **density**?

- Mass per unit of volume
- Represented with the small Greek letter rho: ρ
- Units are kg/m^3

Basic Concepts Related to Kinetics

What is **specific weight**?

- Weight per unit of volume
- Represented with the Greek letter gamma: γ
- Units are N/m^3

Basic Concepts Related to Kinetics

A shot and softball have similar volumes, but the shot has greater mass and therefore greater density. Because weight is proportional to mass, the shot also has greater weight and therefore greater specific weight.

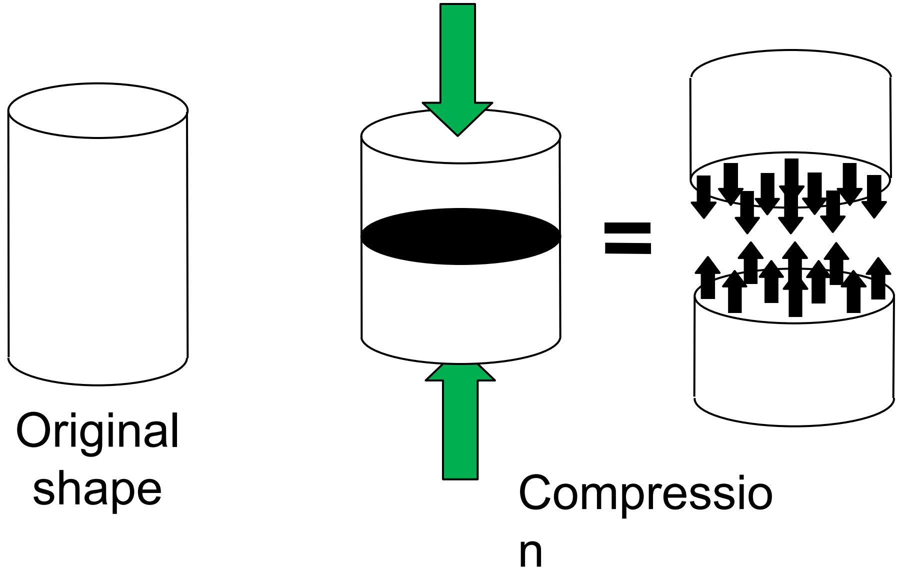
Basic Concepts Related to Kinetics

What is **impulse**?

- The product of force and the time over which the force acts (Ft)
- Units are Ns

Basic Concepts Related to Kinetics

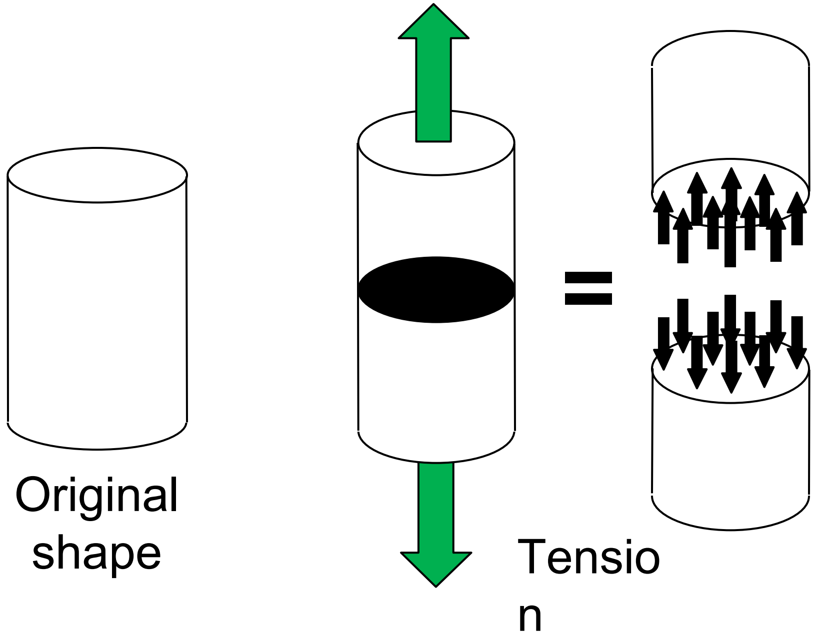
What is **compression**?



(Pressing or squeezing force directed axially through a body)

Basic Concepts Related to Kinetics

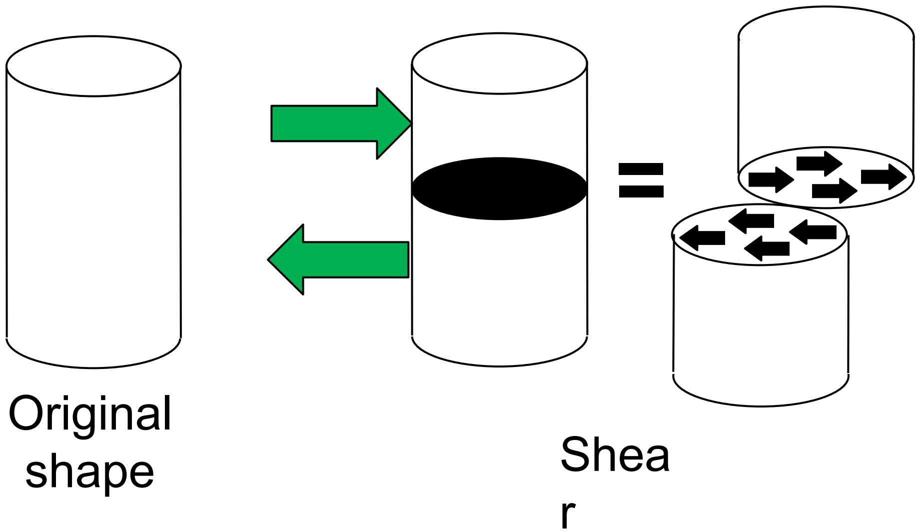
What is **tension**?



(Pulling or stretching force directed axially through a body)

Basic Concepts Related to Kinetics

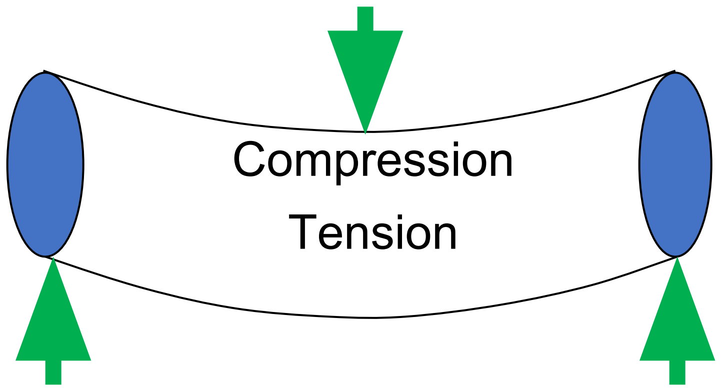
What is **shear**?



(Force directed parallel to a surface)

Basic Concepts Related to Kinetics

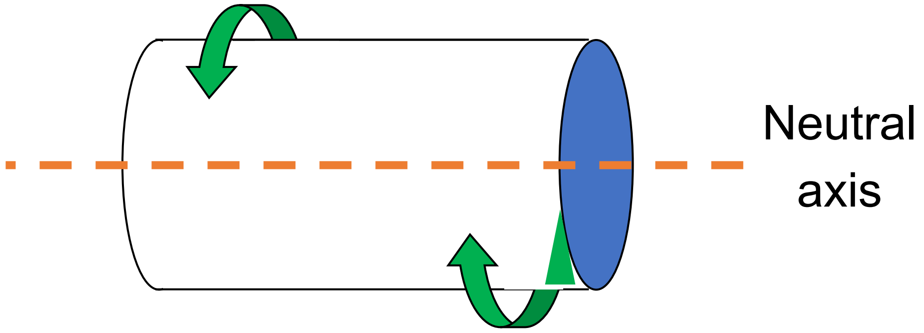
What is **bending**?



(Asymmetric loading that produces tension on one side of a body's longitudinal axis and compression on the other side)

Basic Concepts Related to Kinetics

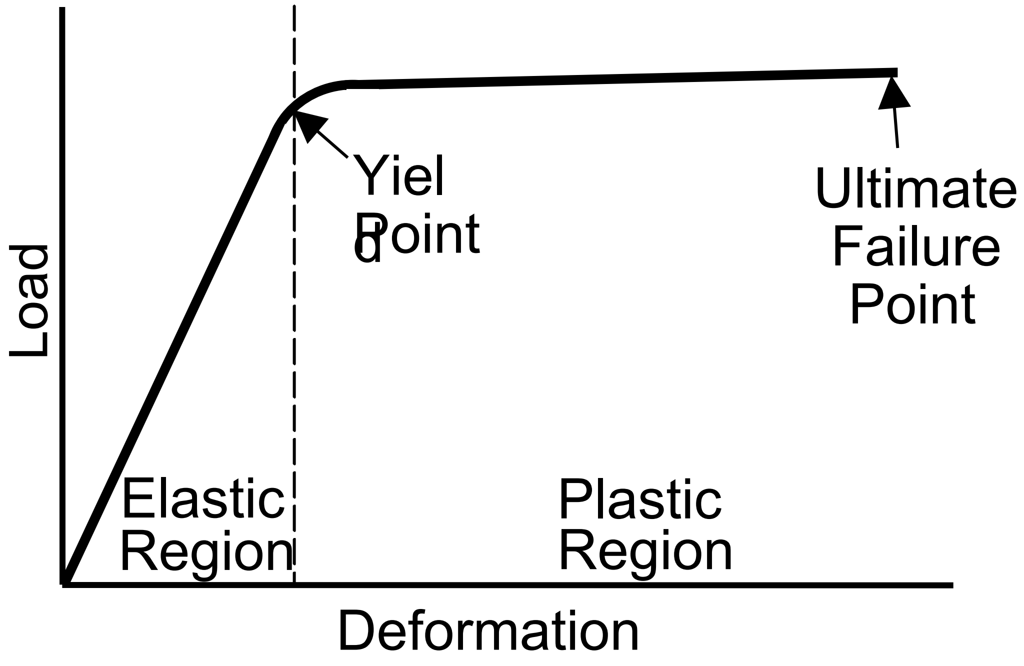
What is **torsion**?



(Load producing twisting of a body around its longitudinal axis)

Basic Concepts Related to Kinetics

What is **deformation**?



(Change in shape)

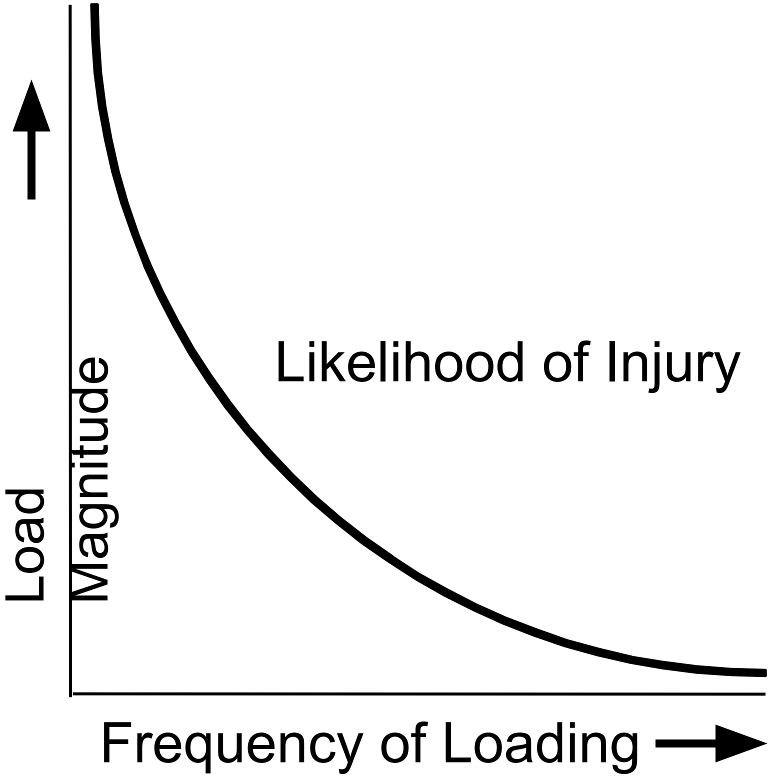
Basic Concepts Related to Kinetics

What are **repetitive** and **acute** loading?

- Repetitive: repeated application of a subacute load that is usually of relatively low magnitude
- Acute: application of a single force of sufficient magnitude to cause injury to a biological tissue

Basic Concepts Related to Kinetics

Repetitive vs. acute loading

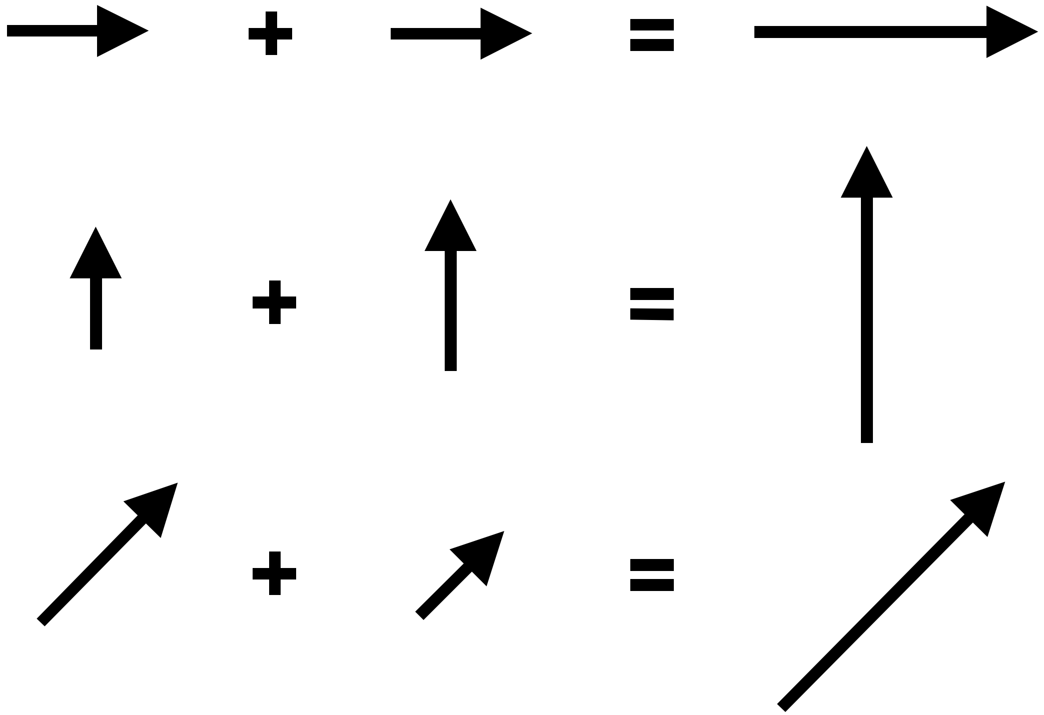


Vector Algebra

What is vector composition?

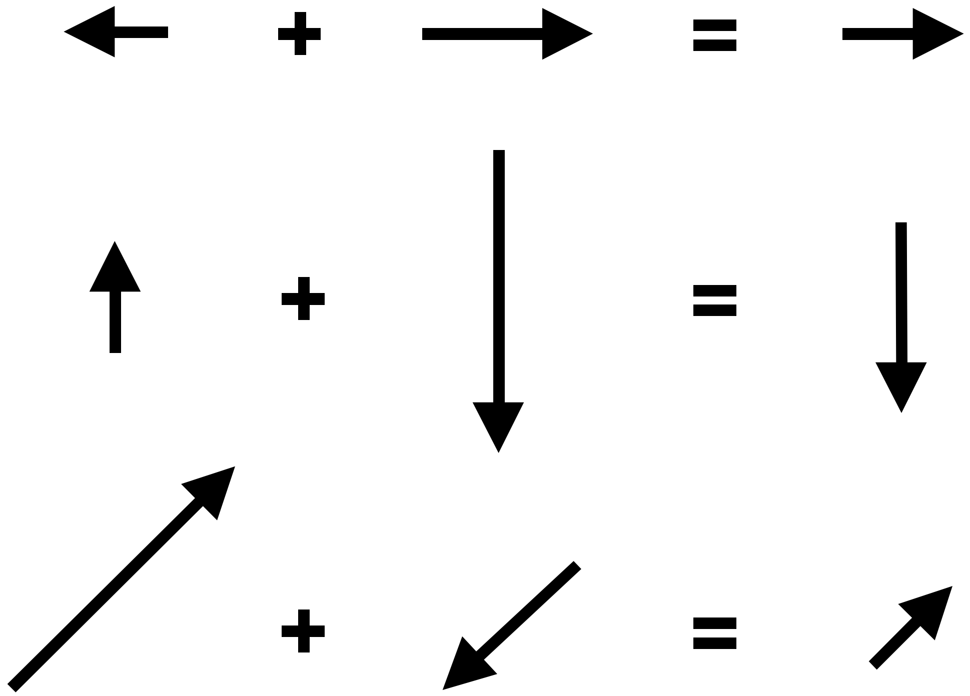
(Process of determining a single vector from two or more vectors by vector addition)

Vector Algebra



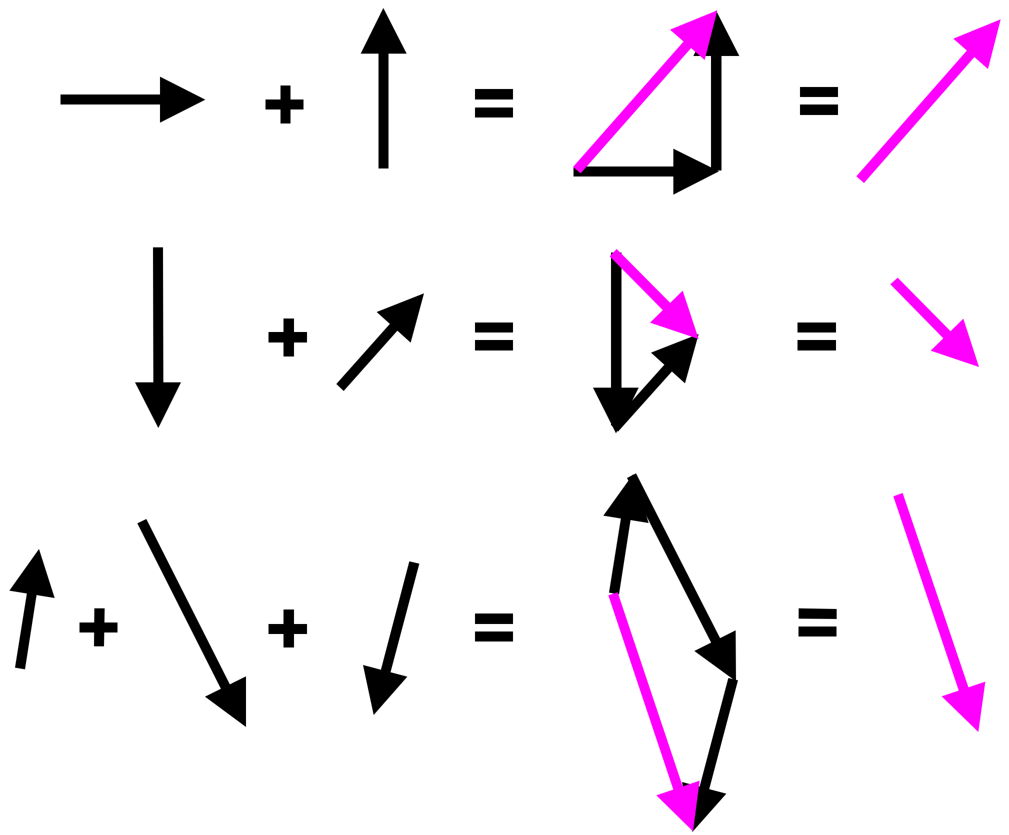
The composition of vectors with the same direction requires adding their magnitudes.

Vector Algebra



The composition of vectors with the opposite directions requires subtracting their magnitudes.

Vector Algebra



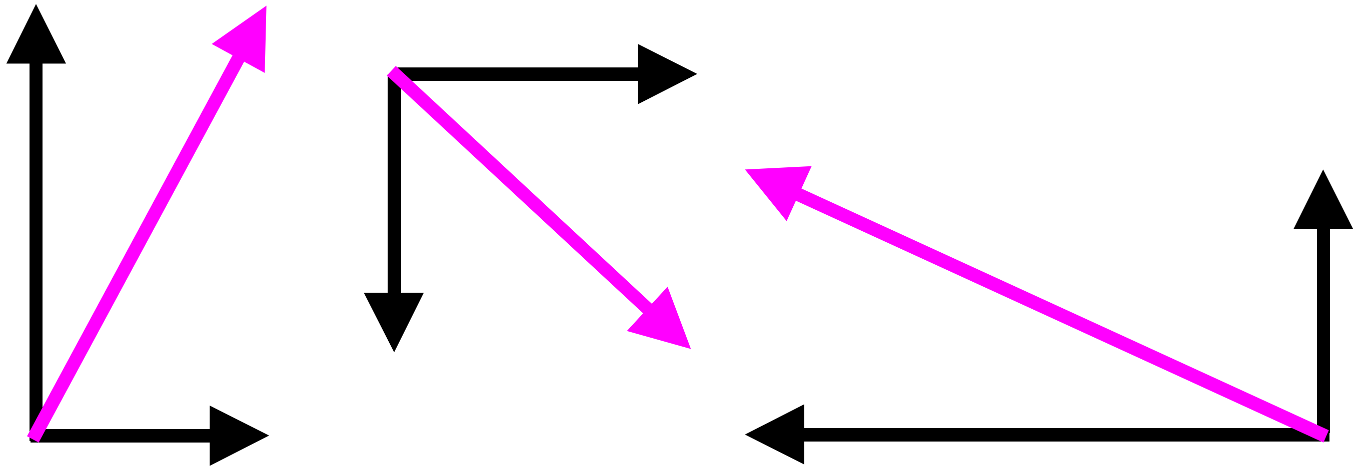
The *tip-to-tail* method of vector composition.

Vector Algebra

What is **vector resolution**?

(Operation that replaces a single vector with two perpendicular vectors such that the vector composition of the two perpendicular vectors yields the original vector)

Vector Algebra



Vectors may be resolved into perpendicular components. The vector composition of each pair of components yields the original vector.

Angular Kinetics of Human Movement

By

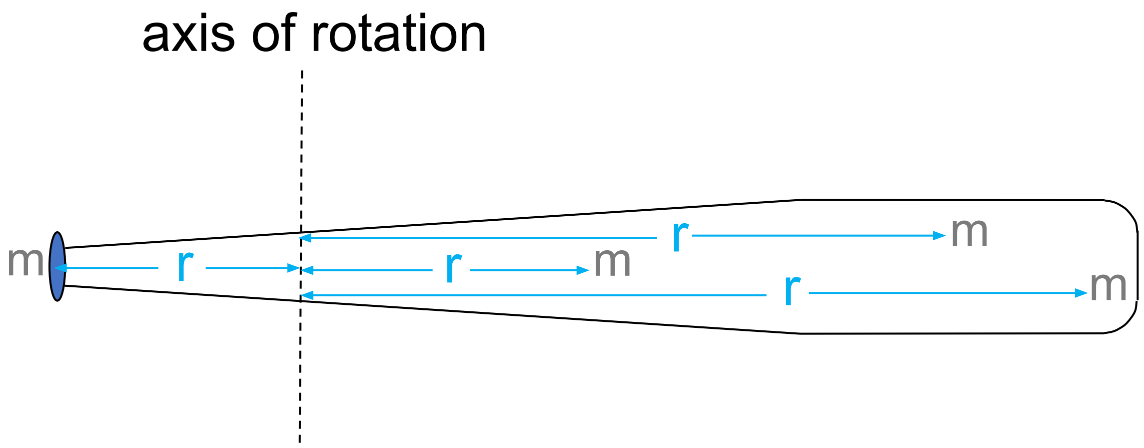
Dr.P.Arunkumar

Resistance to Angular Acceleration

Moment of Inertia

- the inertial property for rotating bodies
- represents resistance to angular acceleration
- based on both mass and the distance the mass is distributed from the axis of rotation

Resistance to Angular Acceleration



Moment of inertia is the sum of the products of each particle's **mass (m)** and the radius of rotation (r) for that particle squared. $I = \Sigma mr^2$

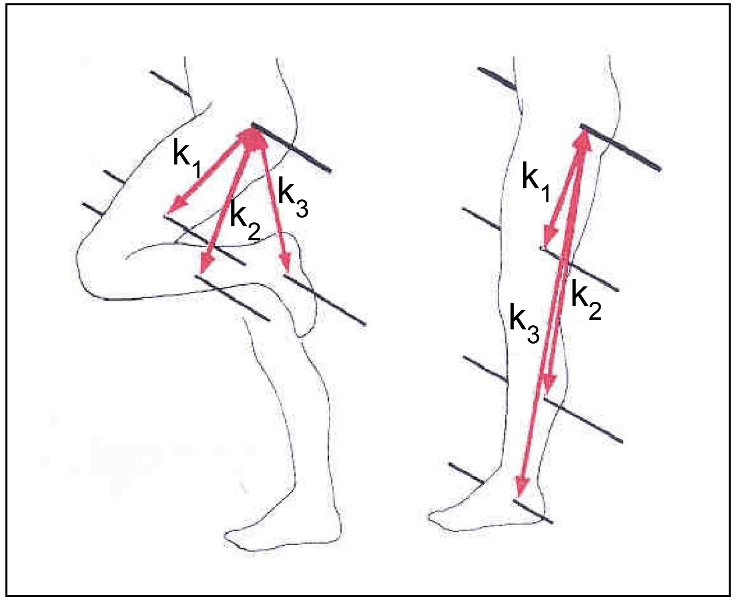
Resistance to Angular Acceleration

Radius of Gyration

- **distance** from the axis of rotation to a point where the body's mass could be concentrated without altering its rotational characteristics
- used as the **index** for mass distribution for calculating moment of inertia:

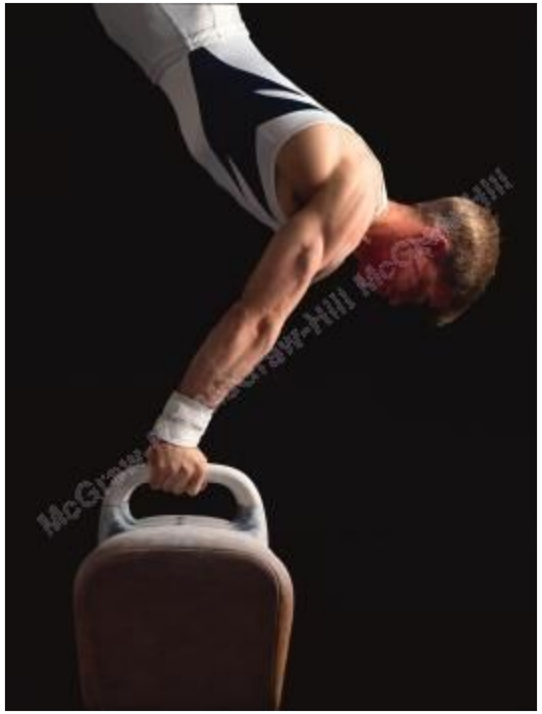
$$I = mk^2$$

Resistance to Angular Acceleration



Knee angle affects the moment of inertia of the swinging leg with respect to the hip because of changes in the radius of gyration for the lower leg (k_2) and foot (k_3).

Resistance to Angular Acceleration



The ratio of muscular strength (ability to produce torque at a joint) to segmental moments of inertia (resistance to rotation at a joint) is important for performance in gymnastic events.

Angular Momentum

- quantity of angular motion possessed by a body
- measured as the product of moment of inertia and angular velocity:

$$H = I\omega$$

$$H = mk^2\omega$$

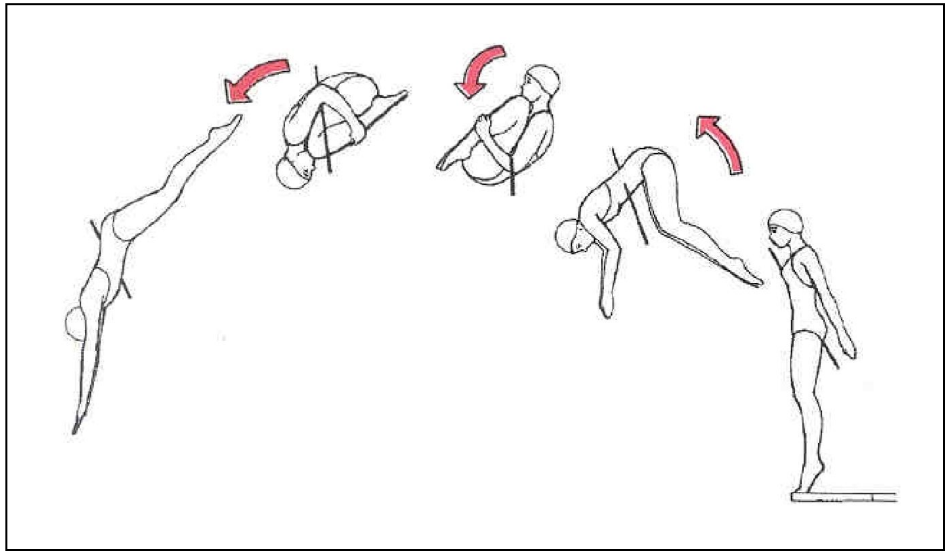
Angular Momentum

Principle of Conservation of Angular Momentum

The total angular momentum of a given system remains constant in the absence of external torques.

$$H_1 = H_2$$
$$(mk^2\omega)_1 = (mk^2\omega)_2$$

Angular Momentum



When angular momentum is conserved, there is a tradeoff between moment of inertia and angular velocity.

(Tuck position = small I, large ω)

(Extended position = large I, small ω)

Angular Momentum

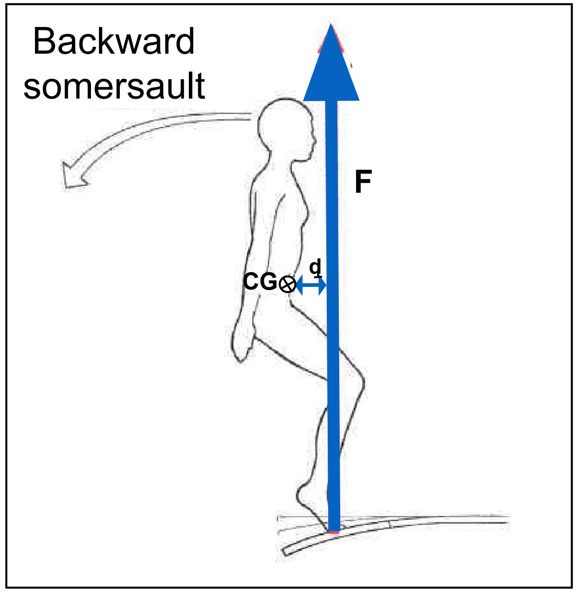
What produces change in angular momentum?

angular impulse - the product of torque and the time interval over which the torque acts:

$$T\ t = \Delta H$$

$$T\ t = (I\omega)^2 - (I\omega)^1$$

Angular Momentum



Springboard reaction force (F) multiplied by its moment arm from the diver's CG (d) creates a torque that generates the angular impulse that produces angular momentum at takeoff.

$$Tt = \Delta H$$

Angular Analogues of Linear Kinematic Quantities

What are the angular equivalents of linear kinematic quantities?

Linear	Angular
mass (m)	moment of inertia ($I = mk^2$)
force (F)	torque ($T = Fd_{\perp}$)
momentum ($M=mv$)	angular momentum ($H=mk^2\omega$)
impulse (Ft)	angular impulse ($Fd_{\perp} t$)

Angular Analogues of Newton's Laws

Angular Law of Inertia

A rotating body will maintain a state of rest or constant rotational motion unless acted on by an external torque that changes the state.

Angular Analogues of Newton's Laws

Angular Law of Acceleration

A net torque causes *angular acceleration* of a body that is:

- of a magnitude proportional to the torque
- in the direction of the torque
- and inversely proportional to the body's moment of inertia

Angular Analogues of Newton's Laws

angular law of acceleration

$$T = I\alpha$$

$$T = mk^2\alpha$$

Angular Analogues of Newton's Laws

Angular Law of Reaction

- For every angular action, there is an equal and opposite angular reaction.
- When one body exerts a torque on a second, the second body exerts a reaction torque that is equal in magnitude and opposite in direction on the first body.

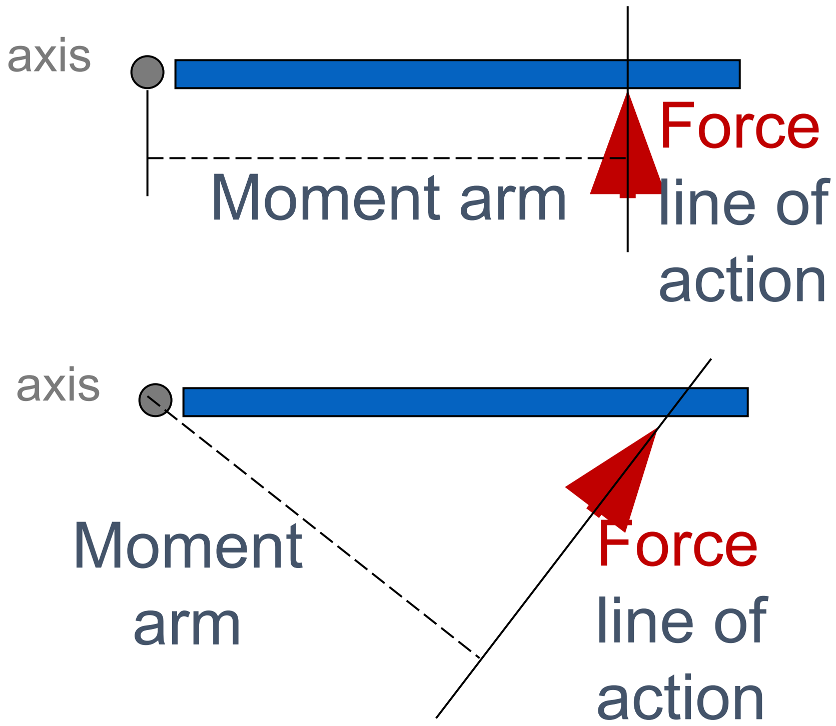
Equilibrium

What is torque?

- the rotary effect of a force about an axis of rotation
- measured as the product of force and the force's moment arm - the shortest (perpendicular) distance between a force's line of action and an axis of rotation

Equilibrium

The moment arm of a force is the perpendicular distance from the force's line of action to the axis of rotation.



Equilibrium

Where do torques occur within the human body?

The product of muscle tension and muscle moment arm produces a torque at the joint crossed by the muscle.

Equilibrium

Skilled pitchers often maximize the length of the moment arm between the hand and total-body axis of rotation during the delivery of a pitch to maximize the effect of the torque produced by the muscles.

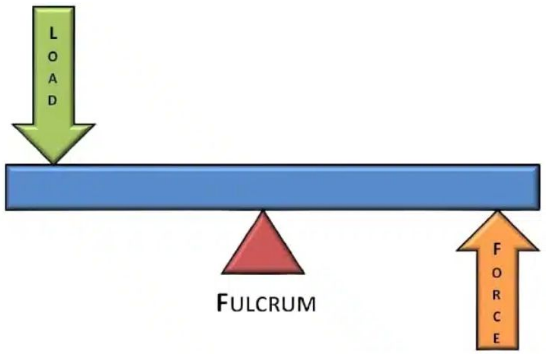
Equilibrium

What is a lever?

- a simple machine consisting of a relatively rigid bar-like body that can be made to rotate about an axis or a fulcrum
- there are first, second, and third class levers

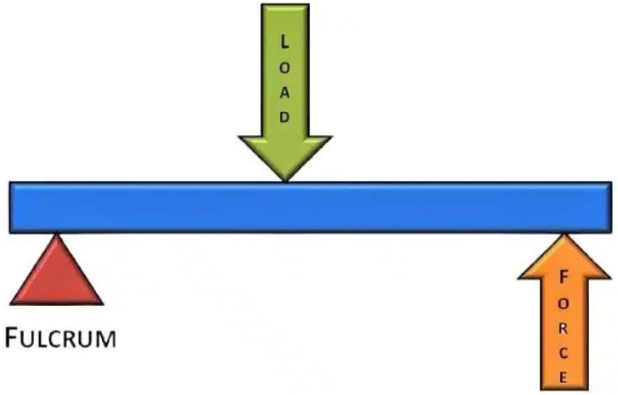
Equilibrium

Relative locations of the applied force (F), the resistance (R), and the fulcrum or axis of rotation determine lever classifications.



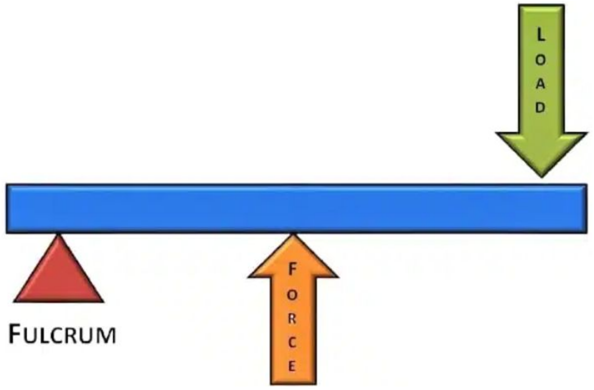
First Class Lever

fulcrum between the force and the load.



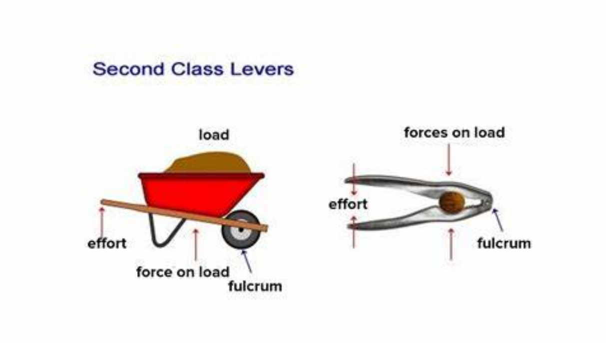
Second Class Lever

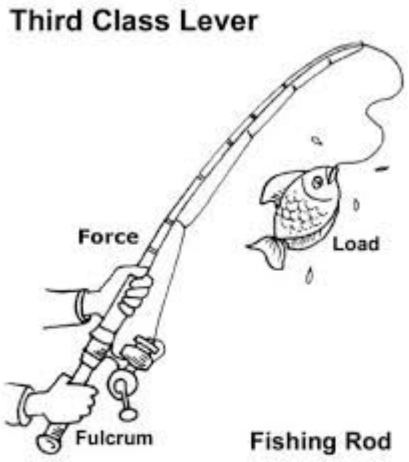
The load is found to be present within the effort that is the force and the fulcrum.



Third Class Lever

The force is found to be present within the load and the fulcrum.



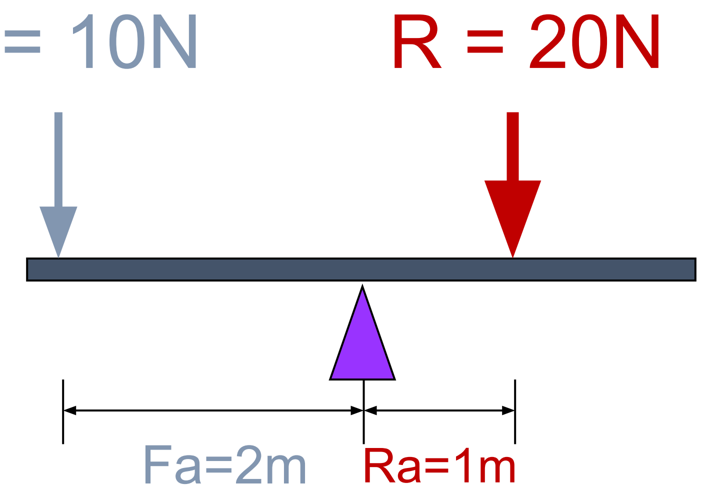


Equilibrium

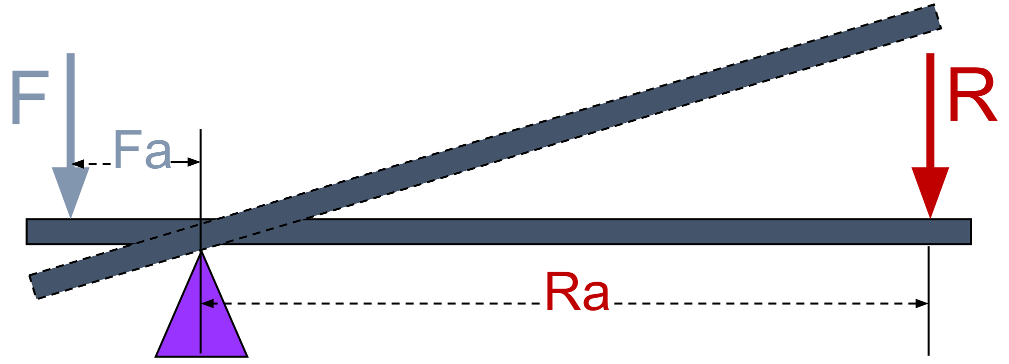
What is mechanical advantage?

the ratio of the moment arm of the force (force arm) to the moment arm of the resistance (resistance arm) for a given lever

Equilibrium



A force can balance a larger resistance when the force arm is longer than the resistance arm.



A force can move a resistance through a large range of motion when the force arm (Fa) is shorter than the resistance arm (Ra).

Equilibrium

What is static equilibrium?

- a motionless state in which there is no net force or net torque acting

- the conditions of static equilibrium are:

$$\Sigma F_v = 0$$
$$\Sigma F_h = 0$$
$$\Sigma T = 0$$

Center of Gravity

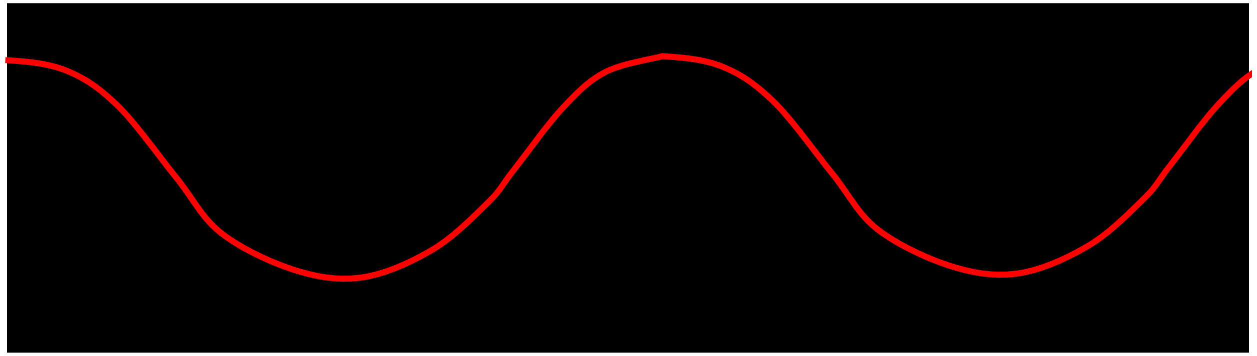
What is the center of gravity?

- the point around which a body's weight is equally balanced in all directions
- also referred to as the center of mass or mass centroid
- (need not be physically located inside of a body)

Center of Gravity

Why is the center of gravity of interest in the study of human biomechanics?

- it serves as an index of total body motion



Path of the center of gravity of a runner.

Center of Gravity

Why is the center of gravity of interest in the study of human biomechanics?

- the body responds to external forces as though all mass were concentrated at the CG
- this is consequently the point at which the weight vector is shown to act in a free body diagram

Stability and Balance

What is stability?

- resistance to disruption of equilibrium

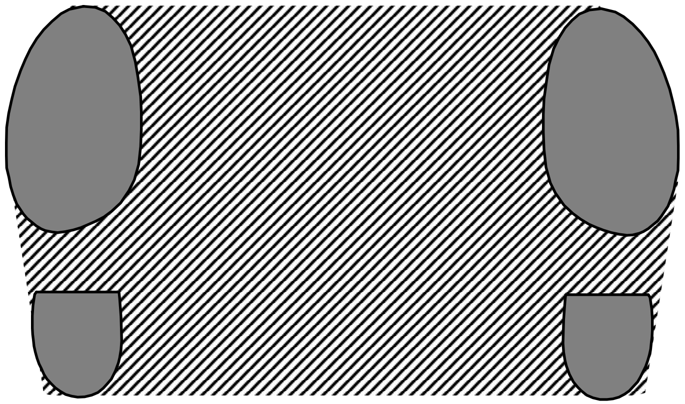
What is balance?

- ability to control equilibrium

Stability and Balance

What is the base of support?

(area bound by the outermost regions of contact between a body and the support surface)



Stability and Balance

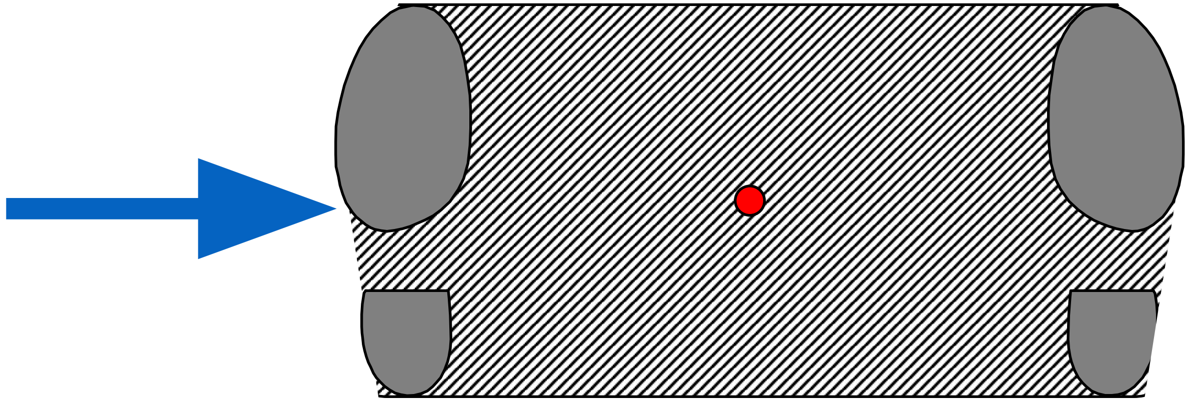
What can increase a body's stability?

- increasing body mass
- increasing friction between the body and the surfaces of contact

Stability and Balance

What can increase a body's stability?

- increasing the size of the base of support in the direction of an external force



Stability and Balance

What can increase a body's stability?

- horizontally positioning the center of gravity near the edge of the base of support on the side of the external force